

Batch-Wise Ensemble Model using Classical and Quantum SVM for Email Phishing Detection

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BATCH-WISE ENSEMBLE MODEL USING CLASSICAL AND
QUANTUM SVM FOR EMAIL PHISHING

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DECLARATION

We hereby declare that the study reported in this thesis entitled as above is our own original work and has not been submitted before anywhere for any degree or other purposes. Further, we certify that the intellectual content of this thesis is the product of our own work and that all the assistance received in preparing this thesis and sources have been acknowledged and cited in the reference Section.

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ABSTRACT

BATCH-WISE ENSEMBLE MODEL USING CLASSICAL AND QUANTUM SVM FOR EMAIL PHISHING

Email phishing is one of the most severe cybersecurity threats in modern communication systems, where attackers use deceptive emails, social engineering techniques and malicious links to steal sensitive information such as login credentials, banking details and personal data. Despite continuous advancements in security mechanisms, traditional rule-based and blacklist-based approaches are still not enough to counter the sophisticated phishing attacks. Although there have been efforts of detecting attacks through classical machine learning algorithms and learning patterns from data, there are challenges on the dependence to feature engineering and generalization across different attacks types. Quantum Machine Learning (QML) is a new paradigm of Machine Learning which has recently shown promise for performing interactions more accurately than classical Machine Learning through the use of quantum kernel methods that use quantum entanglement to map data into high-dimensional Hilbert spaces. This thesis proposes creating a hybrid dataset by modelling real-world phishing behavior attributes along with some of the raw dataset attributes. A hybrid classical-quantum machine learning framework, combining Support Vector Machine (SVM) and Quantum Support Vector Machine (QSVM), in a single ensemble architecture, is proposed to make use of this data. Classical SVM has good deterministic classification ability, but the ability of separability of QSVM is strengthened by quantum feature mapping using the kernel method in feature space of higher dimension. Both outputs of both models are calibrated to guarantee equal probabilistic interpretation for ensemble learning and for comparing both models in a common framework. There are comprehensive experiments carried out with statistical evaluation techniques along with ablation tests to measure the system to see the contribution of the individual components. The experiments indicate that compared to the classical SVM, QSVM can achieve 80% fewer false negatives and with similar precision. The stacking ensemble outperforms both of the base models with an overall accuracy of 99.04% and resolves the precision – recall trade-off problem that happens between the two base models. Overall, this work highlights the effectiveness of combining multi-source feature engineering with classical–quantum hybrid learning to enhance phishing email detection systems and improve cybersecurity resilience.

সারসংক্ষেপ

BATCH-WISE ENSEMBLE MODEL USING CLASSICAL AND QUANTUM SVM FOR EMAIL PHISHING

ইমেইল ফিশিং আধুনিক যোগাযোগ ব্যবস্থার অন্যতম গুরুতর সাইবার নিরাপত্তা হুমকি, যেখানে আক্রমণকারীরা প্রতারণামূলক ইমেইল, সামাজিক প্রকৌশল কৌশল এবং ক্ষতিকর লিংকের মাধ্যমে লগইন তথ্য, ব্যাংকিং তথ্য ও ব্যক্তিগত উপাত্তের মতো সংবেদনশীল তথ্য চুরি করে। নিরাপত্তা ব্যবস্থার ধারাবাহিক উন্নতির পরও, প্রচলিত rule-based এবং blacklist-based পদ্ধতিগুলো উন্নত ও জটিল ফিশিং আক্রমণ প্রতিরোধে এখনো পর্যাপ্ত নয়। যদিও Classical Machine Learning-ভিত্তিক বিভিন্ন পদ্ধতির মাধ্যমে ডেটা থেকে নিদর্শন শিখে ফিশিং শনাক্ত করার প্রচেষ্টা হয়েছে, তবুও বৈশিষ্ট্য নির্মাণের উপর নির্ভরশীলতা এবং বিভিন্ন ধরনের আক্রমণের ক্ষেত্রে সাধারণীকরণ সক্ষমতার সীমাবদ্ধতা রয়ে গেছে। Quantum Machine Learning (QML) হলো Machine Learning-এর একটি নতুন ধারা, যা quantum kernel method ব্যবহার করে quantum entanglement-এর মাধ্যমে ডেটাকে উচ্চ-মাত্রিক Hilbert space-এ রূপান্তর করে এবং Classical Machine Learning-এর তুলনায় অধিক কার্যকর শ্রেণিবিন্যাসের সম্ভাবনা প্রদর্শন করেছে। এই গবেষণায় বাস্তব ফিশিং আচরণের বৈশিষ্ট্য এবং মূল ডেটাসেটের কিছু বৈশিষ্ট্য একত্রিত করে একটি hybrid dataset তৈরি করা হয়েছে। এই ডেটার সর্বোত্তম ব্যবহার নিশ্চিত করার জন্য Support Vector Machine (SVM) এবং Quantum Support Vector Machine (QSVM)-কে সমন্বিত করে একটি hybrid classical-quantum machine learning framework প্রস্তাব করা হয়েছে, যা একটি একক ensemble architecture-এর অধীনে কাজ করে। Classical SVM নির্ভরযোগ্য ও নির্ধারিত শ্রেণিবিন্যাস ক্ষমতা প্রদান করে, অন্যদিকে QSVM উচ্চ-মাত্রিক বৈশিষ্ট্য স্থানে quantum feature mapping-এর মাধ্যমে অধিকতর শক্তিশালী পৃথকীকরণ ক্ষমতা অর্জন করে। Ensemble learning-এর জন্য এবং উভয় মডেলকে একটি অভিন্ন কাঠামোর মধ্যে তুলনা করার উদ্দেশ্যে মডেল দুটির আউটপুটকে সম্ভাব্যতাবিত্তিকভাবে সামঞ্জস্য করা হয়েছে। কর্মদক্ষতা বিশ্লেষণ নির্দেশ করে যে, Classical SVM-এর তুলনায় QSVM প্রায় ৮০% কম False Negative উৎপন্ন করতে সক্ষম হয়েছে এবং একই সঙ্গে সমতুল্য Precision বজায় রেখেছে। এছাড়া, Stacking Ensemble উভয় ভিত্তিমূলক মডেলের তুলনায় উন্নত ফলাফল প্রদান করেছে এবং 99.04% Accuracy অর্জনের পাশাপাশি দুই মডেলের মধ্যে বিদ্যমান Precision-Recall ভারসাম্য সমস্যাও কার্যকরভাবে সমাধান করেছে। সার্বিকভাবে, এই গবেষণা প্রমাণ করে যে বহুমাত্রিক উৎস থেকে প্রাপ্ত বৈশিষ্ট্য নির্মাণ এবং classical-quantum hybrid learning-এর সমন্বয় ইমেইল ফিশিং শনাক্তকরণ ব্যবস্থার কার্যকারিতা বৃদ্ধি করতে পারে এবং সাইবার নিরাপত্তা ব্যবস্থার স্থিতিস্থাপকতা আরও শক্তিশালী করতে গুরুত্বপূর্ণ ভূমিকা পালন করতে সক্ষম।

LIST OF MAIN NOTATION

<i>ML</i>	Machine Learning
<i>SVM</i>	Support Vector Machine
<i>QML</i>	Quantum Machine Learning
<i>QSVM</i>	Quantum Support Vector Machine
<i>NISQ</i>	Noisy Intermediate-Scale Quantum
<i>RBF</i>	Radial Basis Function
<i>MI</i>	Mutual Information
<i>QSVC</i>	Quantum Support Vector Classifier
<i>TF-IDF</i>	Term Frequency-Inverse Document Frequency
<i>AUC</i>	Area Under Curve
<i>ROC</i>	Receiver Operating Characteristics
<i>CV</i>	Cross-Validation
<i>TP</i>	True Positive
<i>TN</i>	True Negative
<i>FP</i>	False Positive
<i>FN</i>	False Negative
<i>LR</i>	Logistic Regression

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CHAPTER 1

INTRODUCTION

1.1 Motivation and Problem Statement

Email phishing is a form of cyberattack that deceives users into revealing sensitive information such as login credentials, financial data or personal identification by imitating trustworthy entities and sending phishing links via email. Amongst all the methods, attackers prefer attacks by emails because they are difficult to detect.(Irani, Webb, Giffin, & Pu, 2008) Attackers have now equipped themselves with machine learning techniques to generate more convincing emails in order to bypass detection mechanisms.(Yamin, Ullah, Ullah, & Katt, 2021)

Traditional phishing detection methods have relied on blacklists, heuristic rules and signature-based filters, which compare incoming emails to known patterns of malicious content. While effective against previously seen attacks, these approaches struggle to detect novel or obfuscated phishing attempts, leading to high false negatives. (Kyaw, Gutierrez, & Ghobakhlou, 2024) To overcome these limitations, machine learning (ML) models such as Support Vector Machines, Random Forests and Logistic Regression have been applied to analyse patterns in email content, URLs and structural features, improving detection rates over static methods. (Jadhav & Chandre, 2025) However, ML-based methods also face challenges, including the need for extensive feature engineering, large labelled datasets and potential performance degradation when encountering unseen or highly adaptive phishing strategies. (Kytidou, Tsikriki, Drosatos, & Rantos, 2025)

The classical ML methods show some drawbacks in phishing detection which led to

the inclusion of the quantum computing and quantum machine learning (QML). Quantum models such as the Quantum Support Vector Machine uses superposition and entanglement to handle high-dimensional feature spaces and capture complex patterns in phishing emails and URLs. But, ensemble evaluation of quantum models is still very expensive in terms of computation. So, in this thesis, a hybrid classical–quantum machine learning framework is proposed which aims to solve the problem of improving the accuracy of phishing email detection. The proposed framework integrates classical Support Vector Machines (SVM) and Quantum Support Vector Machines (QSVM) within a stacking-based ensemble architecture to improve phishing detection capability, reduce false negatives and provide a more adaptive solution for detecting sophisticated phishing attacks.

1.2 Thesis Objectives

The objectives of this thesis are as follows:

1. To construct a hybrid feature-based phishing dataset suitable for both classical and quantum models.
2. To develop a hybrid ensemble framework integrating SVM and QSVM for phishing email detection.
3. To evaluate and analyze the performance of the proposed hybrid ensemble model in comparison with its individual classical and quantum components.

1.3 Methodological Overview

This thesis proposes a hybrid phishing email detection framework that combines classical machine learning and quantum machine learning techniques within a stacking ensemble architecture. Initially, a hybrid dataset is constructed by integrating phishing email

features with webpage and domain-level attributes obtained from multiple cybersecurity datasets. Additional engineered features related to URL structure, entropy, domain trust, semantic consistency and webpage content are extracted to enhance representation quality.

After preprocessing, the most informative features are selected for quantum model training due to quantum computational constraints.

For classification, two parallel models are implemented: a classical Support Vector Machine (SVM) with a radial basis function kernel and a Quantum Support Vector Machine (QSVM) using a ZZFeatureMap-based quantum kernel. The QSVM embeds classical features into a quantum feature space where complex nonlinear relationships can be modeled through quantum state representations.

To reduce computational overhead and improve model diversity, a batch-wise training strategy is adopted. Predictions from both SVM and QSVM models are then calibrated and aggregated through a stacking ensemble approach using Logistic Regression as the meta-classifier. Finally, the framework is evaluated using multiple classification metrics, statistical significance testing and ablation studies to assess its effectiveness and reliability.

1.4 Research Contribution

The main contributions of this thesis can be summarised as follows:

- **Development of a Hybrid Phishing Dataset:** Hybrid phishing dataset was constructed by combining email metadata with URL, domain and website related data. In order to, draw useful structural, lexical, entropy-based and trust based attributes, advanced feature engineering techniques were applied, giving us more complex

features for phishing for machine learning and quantum machine learning models.

- **Design of a Classical–Quantum Phishing Detection Framework:** A hybrid detection framework was designed, which involved classical SVM along with quantum SVM for email phishing classification. The framework allows for a systematic exploration of the quantum enhanced methods in a cyber security scenario.
- **Implementation and Comparative Evaluation of Classical and Quantum Models:** The classical and quantum models, namely Classical Support Vector Machine (SVM) and Quantum Support Vector Machine (QSVM), were implemented and compared with the same experimental setup, providing a detailed performance comparison.
- **Development of an Ensemble-Based Detection Model:** An ensemble classifier combining the strengths of individual models was proposed. Experimental results show that the ensemble approach gives the best performance in the phishing detection compared to individual SVM and QSVM classifiers.
- **Comprehensive Validation through Statistical and Ablation Analysis:** Statistical significance testing and ablation studies were performed to test the reliability of the results, to quantify the contribution of the various groups of features and model components, providing more insights into the effectiveness of the proposed model framework.

1.5 Scope of The Thesis

This thesis focuses specifically on email phishing detection using hybrid classical and quantum machine learning techniques. The research primarily investigates the effectiveness of QSVM and classical SVM within a stacking-based ensemble framework for binary

classification of phishing and legitimate emails.

The scope of the work includes dataset construction, feature engineering, preprocessing, quantum feature mapping, ensemble learning and performance evaluation. To create realistic phishing behaviour, both email level and webpage level characteristics are included. Ablation studies and statistical analysis are also provided to verify the effectiveness of the models considered and assess the effects of the components of the framework.

The thesis however, is confined to simulation based quantum machine learning instead of running on real quantum hardware, through the use of quantum statevector simulators. Because of the current limitation of Noisy Intermediate-Scale Quantum (NISQ) hardware and computational constraints, the feature dimensionality and dataset size are restricted for practical QSVM implementation.

CHAPTER 2

THEORETICAL BACKGROUND AND LITERATURE REVIEW

2.1 Theoretical Background

2.1.1 Email Phishing

Email phishing is a method of cyber attack where attackers attempt to gain data or information via fake emails. Such e-mails are generally masquerading as genuine email messages from carefully reputable organizations, such as banks, social media platforms, ecommerce services or government agencies. The primary objective of phishing attacks is to exploit human trust and psychological manipulation through social engineering techniques.

The attacker develops misleading text in the email to appear as if coming from a trusted organization through spoofed sender address, domain or a credible email format. The email usually includes malicious links or includes an attachment or urgent request for the user to click on it. As the victim clicks on the malicious link or downloads the attachment, attackers are able to steal credentials and deploy malware or redirect to counterfeit websites.

Today phishing attacks are very sophisticated because of the existence of automated phishing toolkits, Artificial Intelligence and the dynamically generated malicious sites. The detection of phishing emails typically relies on extracting discriminative features. These are analysed by rule-based systems or heuristic methods/machine learning tech-

niques and classified as legitimate or malicious emails.

2.1.2 Machine Learning for Phishing Detection

Machine learning has become one of the most effective approaches for detecting phishing emails, which can learn hidden patterns from large data sets and classify malicious email activities automatically. Traditional rule-based systems use predefined rules like signatures and the words on the URL to determine whether the email is spam. The relationship between email content, URL signatures and words on the URL is analyzed by machine learning models to detect phishing attempts using structures, domain information and webpage characteristics.

Among various machine learning algorithms, Support Vector Machine(SVM) is widely used for phishing detection because of its strong performance in binary classification problems. The basic idea behind SVM is to find an optimal hyperplane that is maximally separated from the phishing samples and legitimate samples. The SVM decision function can be represented as:

$$f(x) = w^T x + b \quad (2.1)$$

where w denotes the weight vector, x represents the feature vector and b is the bias term. For nonlinear phishing patterns, kernel functions are applied to transform data into higher-dimensional feature spaces where classification becomes easier. One commonly used kernel is the Radial Basis Function (RBF) kernel:

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2) \quad (2.2)$$

Despite achieving high detection accuracy, classical machine learning models often

face limitations related to feature engineering complexity, scalability and reduced adaptability against increasingly sophisticated phishing attacks.

2.1.3 Quantum Machine Learning for Phishing Detection

Quantum Machine Learning (QML) integrates quantum computing principles with machine learning techniques to improve computational capability and pattern recognition performance. In phishing detection, QML aims to enhance the classification of complex and highly nonlinear phishing patterns by utilizing quantum properties such as superposition and entanglement. Unlike classical bits, quantum bits or qubits can exist in multiple states simultaneously, enabling quantum systems to process high-dimensional information more efficiently.

A qubit can be mathematically expressed as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \quad (2.3)$$

where α and β are probability amplitudes satisfying the normalization condition.

One of the most prominent QML algorithms used for classification tasks is the Quantum Support Vector Machine (QSVM). QSVM extends the traditional SVM framework by replacing classical kernel computation with quantum kernel estimation. The quantum kernel function is defined as:

$$K(x_i, x_j) = |\langle\phi(x_i)|\phi(x_j)\rangle|^2 \quad (2.4)$$

where $|\phi(x_i)\rangle$ and $|\phi(x_j)\rangle$ represent quantum feature states encoded through quantum circuits. By mapping classical phishing features into quantum feature spaces, QSVM can potentially improve feature separability and classification performance. However,

current quantum systems remain constrained by hardware limitations, noise and limited qubit availability. As a result, modern phishing detection research increasingly focuses on hybrid classical–quantum frameworks that combine classical machine learning with the advanced feature representation capability of quantum models.

2.2 Literature Review

2.2.1 Related Works

According to the literature, classical machine learning methods, given that appropriate feature engineering is used and the dataset size is sufficient, still show a high level of competitiveness. Recent empirical studies consistently identify Support Vector Machines preeminent classical technique on the detection of email phishing, achieving robust performance on large scale datasets. (Murti & Naveen, 2023) . Similarly, another validation study using machine learning methods to implement a phishing email detection mechanism found that SVM showed the best accuracy of 97.6%, which was improved after feature selection (Fares et al., 2024). Research has been conducted on employing Quantum Support Vector Machines (QSVMs) to solve cyber security classification problems in order to enhance the detection capability in high dimensional and complex feature space. A study showed that the QSVM can be used to explore the feature spaces complementarily by quantum feature mapping and quantum kernel alignment (Grossi et al., 2022). The study proposed the idea of the hybrid ensemble learning between classical and quantum model. The authors noted that when quantum and classical classifiers were used, overall decision-making results were better. Their results help to suggest that quantum improved models can help decrease classification errors in cybersecurity applications, without losing the benefits of classical systems in terms of scalability and deployment feasibility. Another systematic review identified that the relationship between the different features

is high dimensional and complex in phishing datasets and that some kind of transformations like quantum feature maps and kernel-based transformations like ZZFeatureMap might improve the separability of the classes and consequently the performance of phishing detection. The review also pointed out that the vast majority of the previous works on phishing detection based on QML are based on shallow quantum circuits and simulator-based experiments that are constrained by the scalability, limited number of available qubits and hardware noise. Moreover, an ensemble of learning and feature engineering were observed to have significant effect in enhancing robust and minimizing misclassification (Alenezi & Alenezi, 2026). Quantum machine learning (QML) in cybersecurity is in the very early stages, often using benchmark data sets such as malware samples, URL lists and intrusion signal logs, instead of full-blown e-mail data sources. A recent cybersecurity study used QSVM for malware classification using PennyLane and the Drebin dataset, highlighting the feasibility of QSVM-style pipelines in security domains (Akter et al., 2023). The researchers achieved an accuracy of 95% in malware classification and protection by QSVM. A comparative study of classical ML used preprocessing and quantum encodings like ZZFeatureMap and ZFeatureMap for malicious URL detection (Eze, Chaudhry, & Jahankhani, 2025). It's supportive of the kernel-based quantum models' (QSVM) capability to be more competitive under the existing hardware conditions. Other kinds of hybrid cybersecurity detection systems based on QSVM have been explored outside of the scope of phishing datasets. The comparisons of detection accuracy, precision and recall and reduction in false positive rate of the proposed hybrid intrusion detection system with feature optimization in IoT security environment with several traditional machine learning algorithms demonstrated the effectiveness of the proposed hybrid system. The authors also noted advantages of employing QSVMs on large scale feature maps and complicated decision boundaries. This reinforces the fact that QSVM could be used as a complementary method to classical SVM techniques in the cybersecurity field where there is a high level of non-linearity among the interactions of features (Elsedimy,

Elhadidy, & Abohashish, 2024). Ensemble machine learning techniques that combine multiple classifiers via majority voting have also been explored for phishing detected to single models (Bari et al., 2025). We found that quantum kernels can outperform classical models on phishing URL datasets (Wei, Sekiya, Nakayama, & Todo, 2025) and a QKD-based Email Validator can confirm the legitimacy of emails by analysing all email traffic for signs of malicious activity (Chandre et al., 2024). The authors constructed the stacking ensemble and bagging ensemble strategies of quantum model and classical model to further enhance the classification performance. Experimental results showed that QSVM consistently outperforms other models in terms of False Positive Rate (FPR) and Precision (Prec) measures and ensemble methods boost the Phishing Detection F1-score. While the models were limited by the availability of training data and by the computational capabilities of today’s quantum hardware which is still in the NISQ era, the paper also proves the viability of quantum kernel methods for use in cybersecurity applications (Ray, Sakunthala, Ajith, & Vinayagamurthy, 2022).

2.2.2 Summary Table

Table 2.1: Comparison of Related Works

Paper	Dataset	Method	Accuracy	Limitations
Machine Learning Algorithms for Phishing Email Detection (Murti & Naveen, 2023)	Kaggle phishing email dataset; 80/20 split, 360,868 train and 90,217 test records	NB, SVM, RF, DT, KNN	RF: 99.45%	No QSVM, no quantum feature mapping and no hybrid classical–quantum ensemble.

2 THEORETICAL BACKGROUND AND LITERATURE REVIEW

Machine Learning Approach for Email Phishing Detection(Fares et al., 2024)	Kaggle dataset with Email Text and Email Type labels	SVM, RF, XG-Boost, TF-IDF, BoW	SVM: 97.612%	No QSVM, no quantum kernel and no stacking ensemble.
Mixed Quantum-Classical Method for Fraud Detection with Quantum Feature Selection(Grossi et al., 2022)	IBM Safer Payments fraud dataset	QSVM, RF, XG-Boost, quantum feature selection, mixed ensemble	QSVM: 78% $\pm 1\%$	Payment fraud domain; reduced dataset due to quantum hardware limitations.
Phishing URL Classification in Cybersecurity Education(Alenezi & Alenezi, 2026)	UCI PhiUSIIL Dataset (235,795 instances, 54 features)	RF, SVM, DT, LR	RF: 93.33%	URL-focused dataset; no QSVM or hybrid ensemble.
Case Study-Based Approach of QML in Cybersecurity(Akter et al., 2023)	Drebin215 Malware Dataset	QSVM using PennyLane	95%	Malware classification rather than phishing email detection.
Quantum-Enhanced Machine Learning for Cybersecurity: Evaluating Malicious URL Detection(Eze et al., 2025)	Malicious URL Dataset	SVM, QSVM, CNN, QCNN	QSVM: 100%	Focused on URL detection rather than email phishing.

2 THEORETICAL BACKGROUND AND LITERATURE REVIEW

A Filter-Based Feature Selection Framework to Detect Phishing URLs Using Stacking Ensemble Machine Learning(Bari et al., 2025)	UCI Phishing Websites, Mendeley Phishing Websites, ISCX-URL2016	Feature Selection, SVM, RF, Gradient Boosting, Stacking Ensemble	Dataset 1: 98.17%, Dataset 2: 96.47%, Dataset 3: 97.61%	URL-focused only; no QSVM or quantum feature mapping.
Classical Ensemble of Quantum-Classical ML Algorithms for Phishing Detection in Ethereum Transaction Networks(Ray et al., 2022)	Ethereum Transaction Network Dataset (3M nodes; 1,165 phishing and 2,972,324 non-phishing accounts)	QSVM, VQC, Classical ML, Quantum Feature Maps, Stacking Ensemble, Bagging Ensemble	Macro F1 = 86%, Phishing F1 = 86%	Blockchain phishing detection rather than email phishing; uses transaction features instead of email/webpage features; not a direct SVM-QSVM ensemble.
Quantum Machine Learning for Phishing Detection: A Systematic Review of Current Techniques, Challenges and Future Directions(Kustiawan & Ghauth, 2026)	EPhiUSIIL 2024, PhishStorm, MalCrawler, Ethereum (Etherscan) and malicious URL datasets.	QSVM, QSVC, VQC, QCNN, QNN, Quantum Feature Maps, QRAC, Amplitude Encoding	100% accuracy on a controlled MalCrawler subset and 92% accuracy on a malicious URL dataset.	Systematic review only; no proposed model; focuses largely on URL phishing and simulator-based QML studies rather than email phishing and SVM-QSVM stacking ensembles.

CHAPTER 3

METHODOLOGY

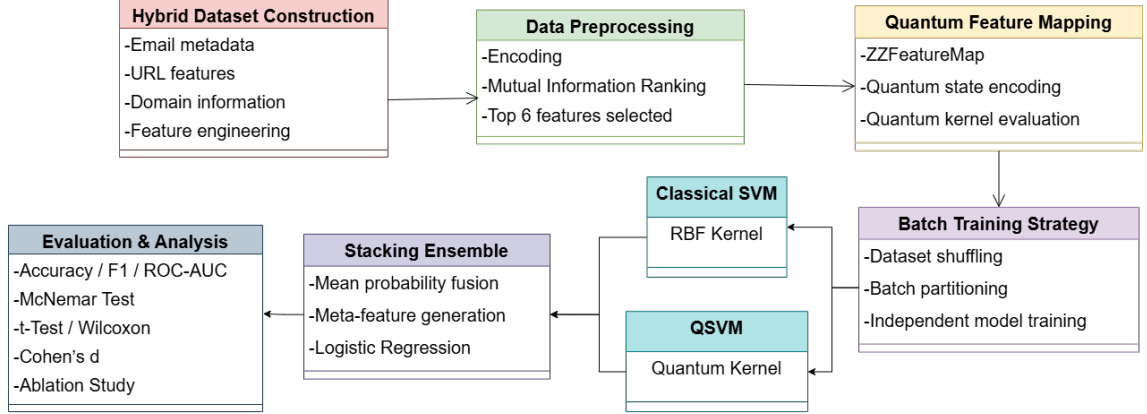


Figure 3.1: Proposed Methodology

3.1 Data Acquisition

From the Phishing Email Dataset (Alam, 2024), only the `CEAS_08.csv` subset was selected, as it provides structured email metadata suitable for feature extraction. A set of email-level features was extracted, including subject length, presence of urgency-related keywords, sender domain, URL length and number of embedded links.

To incorporate website-level characteristics, the extracted email feature dataset was combined with the Malicious and Benign Webpage Dataset (Singh, 2020). The merged dataset was reduced to 8,000 samples, maintaining an equal proportion of malicious and benign instances to prevent class imbalance. Additionally, domain age was extracted for each URL, as it is a strong indicator of malicious activity in phishing attacks.

To align the dataset with a realistic phishing scenario, several attributes were removed, including email URL length, number of links, email label and unnamed web columns. This design

choice ensures that the email body and content features remain unchanged, while the phishing links embedded within emails are assumed to originate from the malicious or benign webpage dataset.

3.2 Hybrid Dataset Construction

A feature engineering pipeline was applied to the base dataset to generate additional discriminative features. This hybridization combines raw dataset attributes with features inspired by real-world phishing behavior. The final hybrid dataset therefore consists of both original attributes and newly extracted features, allowing classical and quantum machine learning models to learn richer representations.

- **Structural and URL-based Features:** Structural characteristics of URLs, including special character counts (`.`, `-`, `/`, `&`, `\_`), number of subdomains, presence of numeric characters in the domain, suspicious top-level domains and IP-based URLs were extracted. A JavaScript obfuscation ratio was additionally computed as the length of obfuscated JavaScript relative to the total JavaScript content:

$$\text{js_obfuscation_ratio} = \frac{\text{web_js_obf_len}}{\text{web_js_len} + 1}$$

- **Entropy-based Features:** Shannon entropy was computed for URLs, domain names and webpage content to quantify randomness and obfuscation:

$$H = - \sum_i p_i \log_2 p_i$$

Higher entropy values indicate automatically generated or maliciously manipulated phishing resources.

- **Domain Trust Score:** A composite domain trust score was calculated using domain age,

HTTPS usage, WHOIS completeness and TLD reputation, yielding a value in $[0, 1]$:

$$\text{trust} = s_{\text{age}} + 0.2 \cdot \mathbf{1}[\text{WHOIS complete}] + 0.2 \cdot \mathbf{1}[\text{HTTPS}] + s_{\text{TLD}}$$

where $s_{\text{age}} \in \{0, 0.1, 0.2, 0.3, 0.4\}$ based on domain age thresholds and $s_{\text{TLD}} \in \{-0.1, 0, 0.2\}$ based on TLD reputation.

- **Email–URL Consistency Features:** Domain matching and similarity metrics between email sender domains and embedded URLs were introduced to identify impersonation attempts. An exact binary match flag and a tiered Jaccard similarity score were both computed.
- **Web Content Analysis Features:** Webpage content was analyzed for HTML form, input and script tags, along with the frequency of phishing-related keywords such as *verify*, *urgent*, *confirm* and *password*. These signs are typical examples of social engineering techniques employed in a phishing attack.
- **Semantic and Brand Consistency Features:** Semantic coherence between email text, URL text and webpage content was measured using TF–IDF vectorization and pairwise cosine similarity. Consistency scores were calculated by comparing known brand key words within the email domain and content to see if brand impersonation behaviors were occurring.

The hybrid dataset presented in this study comprises 38 features, including 14 original features obtained from various sources and 24 features engineered systematically. In order to have a fair and consistent experimental comparison between classical SVM and QSVM models, this dataset is used uniformly for both models.

3.3 Data Preprocessing

3.3.1 Data Cleaning and Missing Value Handling

An initial inspection revealed missing values in two attributes: `email_from_domain` and `domain_age`.

- **email_from_domain:** There were 178 records with missing sender domain (2.2% of the data). As this attribute is a key structural feature for email–URL consistency analysis and its omission is unlikely to introduce systematic bias at this small proportion, the affected rows were removed. This reduced the dataset to 7,822 samples.
- **domain_age:** A total of 2,332 records (29.2% of the dataset) had missing domain age values. Removing these rows would have caused significant data loss and potential sampling bias. Crucially, missingness in domain age is itself informative, unknown or unresolvable domain age frequently indicates a newly registered or deliberately hidden domain, both of which are associated with phishing activity. To preserve this signal, a binary missingness indicator was introduced before imputation. Imputing with zero is a deliberate domain informed choice: a value of zero explicitly encodes the interpretation that the domain is newly observed or unregistered, rather than masking the absence of information with an arbitrary statistic such as the mean.

After handling missing values, the dataset contained 7,822 complete samples with no remaining null entries.

3.3.2 Encoding of Categorical Features

Since Support Vector Machines and quantum kernels require numerical inputs, all categorical attributes were transformed using appropriate encoding strategies:

- **Email domain** (`email_from_domain`) was encoded using frequency encoding, where

each domain is replaced by its normalized occurrence frequency across the training corpus. This preserves relative domain prevalence without introducing high-cardinality one-hot vectors.

- **Geographical location** (`web_geo_loc`) was also encoded using frequency encoding to capture regional distribution patterns without introducing high dimensionality.
- **Top-level domain** (`web_tld`) was mapped to a binary trust score, assigning a value of 1.0 to widely trusted TLDs (`.com`, `.org`, `.net`, `.edu`, `.gov`) and 0.0 otherwise.
- **WHOIS status** (`web_who_is`) and **HTTPS availability** (`web_https`) were encoded as binary values (1 = complete/yes, 0 = incomplete/no).

Highly noisy and unstructured attributes (`web_url`, `web_ip_add` and `web_content`) were removed after feature extraction, as their discriminative information had already been captured through the engineered features. After encoding, the dataset consisted of 36 numerical features, including the missingness indicator.

3.3.3 Train–Test Splitting and Class Distribution

The processed dataset was divided into training and testing subsets using an 80:20 split, while maintaining class proportions through stratified sampling.

- **Training set:** 6,257 samples
- **Test set:** 1,565 samples

Both subsets exhibited nearly equal distributions of phishing and legitimate samples, confirming that no class imbalance was present and therefore no resampling techniques were required.

Table 3.1: Summary of the Final Hybrid Phishing Dataset

Component	Description
Initial Email Dataset	CEAS_08 subset from Phishing Email Dataset (Alam, 2024)
Webpage Dataset	Malicious and Benign Webpage Dataset (Singh, 2020)
Merged Dataset Size	8,000 samples (balanced: 50% phishing, 50% benign)
Final Dataset Size	7,822 samples after preprocessing and cleaning
Original Features	14 features (email + webpage attributes)
Engineered Features	24 features (structural, entropy, trust, semantic, consistency)
Number of Features (Final)	36 numerical features(including missingness-indicator)
Class Distribution	Balanced (stratified sampling maintained)
Train–Test Split	80:20 stratified split
Training Set Size	6,257 samples
Test Set Size	1,565 samples

3.3.4 Feature Scaling

All numerical features were normalized to the range $[0, 1]$ using Min–Max scaling. Feature scaling is critical for SVM-based classifiers, as kernel functions are sensitive to feature magnitudes. The scaler was fitted exclusively on the training set and subsequently applied to the test set to prevent data leakage.

3.3.5 Feature Selection Using Mutual Information

Dimensionality reduction was necessary prior to quantum encoding due to the qubit overhead imposed by the `ZZFeatureMap`, which allocates one qubit per input feature. Mutual Information (MI) was selected as the ranking criterion over alternative methods for the following reasons:

- **Model-agnostic non-linearity:** Unlike Pearson correlation or ANOVA F-score, MI captures both linear and non-linear statistical dependencies between each feature and the target label, making it better suited to the mixed-type, engineered features present in the hybrid dataset.

- **No distributional assumptions:** Methods such as chi-squared testing assume specific data distributions and are restricted to non-negative inputs. MI imposes no such constraints and is applicable to the full range of continuous and binary features in this dataset.
- **Leakage-free estimation:** The feature selection was only carried out on the training set and the resulting ranking was applied to the test set without re-fitting. No information from the test set was used in feature selection.
- **Compatibility with quantum constraints:** Due to the fact that MI produces a global scalar score of importance per feature, it can be used to provide a principled and reproducible reason for selecting out the smallest subset of features which retains the most discriminative information directly related to the objective of minimising the number of qubits for NISQ constraints.

MI scores were calculated on the scaled training set and all 36 features were ranked in decreasing order of informativeness.

The top six features — `web_js_len`, `js_obfuscation_ratio`, `web_js_obf_len`, `content_num_scripts`, `content_entropy` and `domain_trust_score` — were retained for model training.

Although reducing 36 features to six may appear aggressive, the chosen `ZZFeatureMap` encodes each feature into a separate qubit, causing qubit requirements to scale directly with feature dimensionality. Under classical simulation and NISQ constraints, this limits the feasible input size. MI therefore preserves the most informative features while ensuring computational feasibility, rather than performing arbitrary elimination. Figure 3.2 visualises the MI scores of the top 15 features. The red dashed line marks the selection cutoff at $k = 6$, separating the retained features (above) from those excluded due to quantum resource constraints (below).

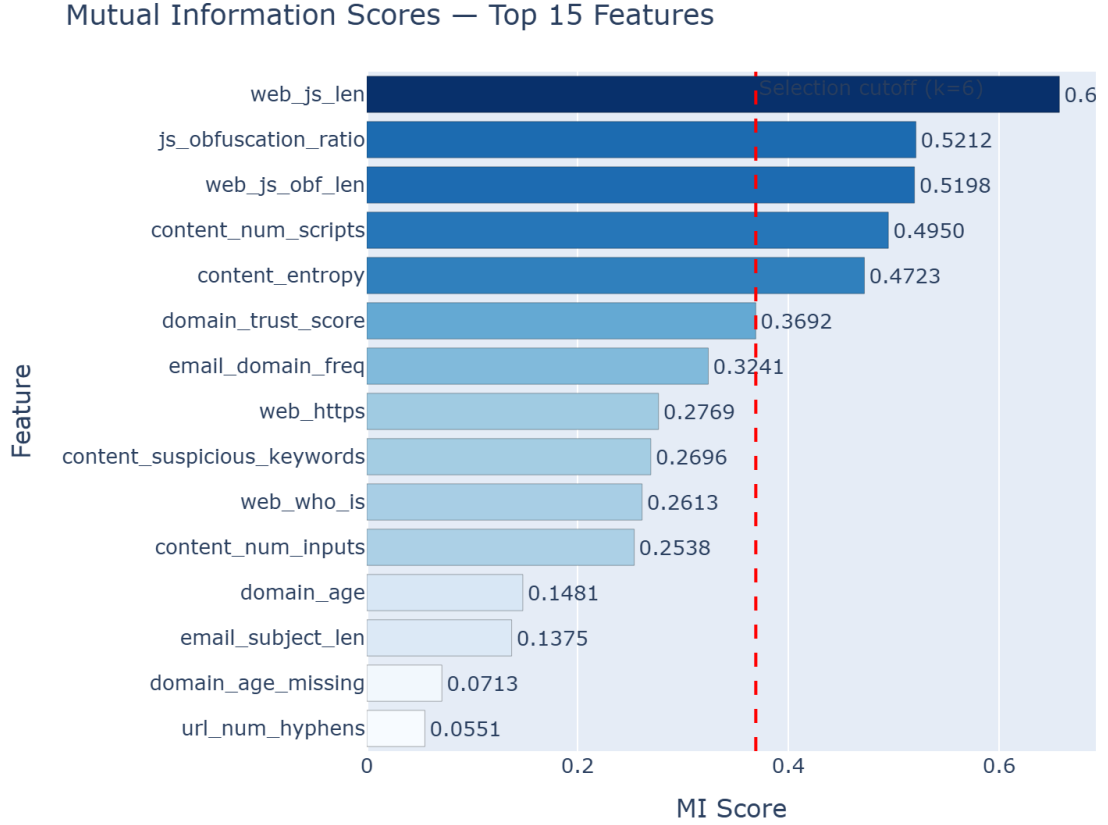


Figure 3.2: Mutual Information scores for the top 15 features, ranked in descending order

The selection of exactly six features was driven by the need to balance predictive performance and computational feasibility. Preliminary experiments showed that six features provided the best trade-off between classification accuracy and quantum simulation cost. Increasing the number of features would substantially increase quantum kernel computation time due to the exponential growth in simulation complexity. Therefore, six features were chosen as the highest feature count that remained practical within the available hardware constraints while maintaining strong classification performance.

3.4 Quantum Feature Mapping and Kernel Evaluation

To incorporate quantum-enhanced learning, classical input vectors were embedded into a quantum feature space using a parameterised quantum circuit. Prior to encoding, all selected features were

rescaled by a factor of π , mapping the normalised range $[0, 1]$ to the rotation range $[0, \pi]$. This rescaling ensures that each feature value is expressed as a meaningful rotation angle on a qubit's Bloch sphere, which is a prerequisite for effective angle encoding in parameterised quantum circuits.

3.4.1 ZZFeatureMap Circuit

The `ZZFeatureMap` from Qiskit's circuit library was employed to encode each rescaled feature vector into a corresponding quantum state. The circuit was configured with the following parameters:

```
feature_map = ZZFeatureMap(
    feature_dimension = 6,    # one qubit per selected feature
    reps              = 1,    # one repetition of the encoding layer
    entanglement      = "linear" # nearest-neighbour entanglement
)
```

The encoding process proceeds in three stages. First, Hadamard gates are applied to all qubits to initialise a uniform superposition state:

$$|0\rangle^{\otimes n} \xrightarrow{H^{\otimes n}} \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle$$

Second, parameterised single-qubit phase rotations $R_Z(\phi_i)$ are applied, embedding each feature value x_i directly into the phase of the corresponding qubit. Third, two-qubit ZZ interaction gates are applied between neighbouring qubit pairs under the linear entanglement scheme, introducing cross-feature correlations of the form $R_{ZZ}(\phi_i \phi_j)$. This entanglement enables the circuit to represent interactions between features that would be inaccessible to classical linear kernels. The resulting quantum circuit for the six-feature configuration is illustrated in Figure 3.3.

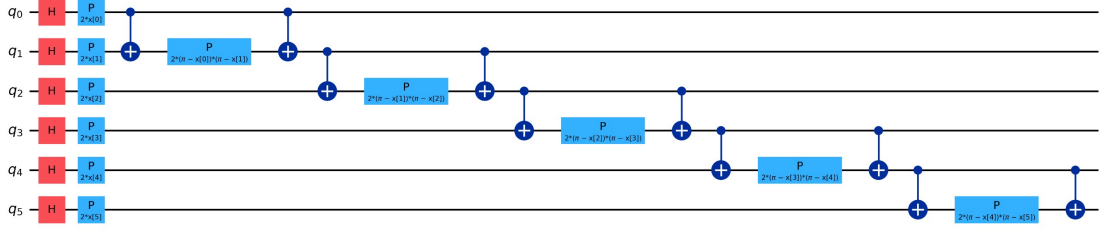


Figure 3.3: Quantum feature encoding circuit using ZZFeatureMap with 6 qubits, 1 repetition and linear entanglement.

3.4.2 Quantum Kernel

The similarity between two data points $\mathbf{x}$ and $\mathbf{x}'$ is quantified through a quantum kernel defined as the squared fidelity between their corresponding quantum states:

$$\kappa(\mathbf{x}, \mathbf{x}') = |\langle \phi(\mathbf{x}) | \phi(\mathbf{x}') \rangle|^2$$

where $|\phi(\mathbf{x})\rangle$ denotes the quantum state produced by the ZZFeatureMap for input $\mathbf{x}$. This kernel effectively measures the overlap between states in a high-dimensional Hilbert space, allowing the model to capture non-linear relationships between features. The kernel was instantiated using Qiskit's FidelityQuantumKernel with the default Sampler primitive:

```
quantum_kernel = FidelityQuantumKernel(feature_map=feature_map)
```

The Sampler primitive evaluates kernel entries by estimating the transition amplitude between two quantum states through circuit sampling, providing an exact simulation-based computation of quantum fidelity under ideal (noise-free) conditions. Specifically, the default Sampler from `qiskit.primitives` performs this evaluation using exact statevector simulation internally, meaning kernel values are computed analytically rather than through stochastic shot sampling. This ensures that all quantum kernel entries are free from sampling noise, making the fidelity estimates exact under the ideal circuit model.

3.5 Batch-wise Training Strategy

One of the major drawback of kernel-based quantum models is the quadratic computational complexity to construct a kernel matrix. For a training set of N samples, computing the full kernel matrix requires $\mathcal{O}(N^2)$ circuit evaluations, making direct training on the full 6,257-sample set computationally infeasible under simulation.

Computational Complexity Analysis

To formally justify the batch-wise strategy, a brief complexity analysis is provided. Let N denote the number of training samples and d the number of selected features (qubits). In this work, $N = 6,257$ and $d = 6$.

For a classical SVM with an RBF kernel, kernel matrix computation scales as $\mathcal{O}(N^2d)$, with training complexity up to $\mathcal{O}(N^3)$, which remains feasible in practice due to optimised solvers.

In contrast, QSVM kernel computation requires simulating a d -qubit quantum circuit for each kernel entry. On a classical simulator, representing a d -qubit state requires 2^d complex amplitudes, incurring exponential memory overhead of $\mathcal{O}(2^d)$ per circuit evaluation. The total cost of constructing the full kernel matrix therefore becomes:

$$\mathcal{O}(N^2 \cdot 2^d)$$

Substituting the experimental values gives $\mathcal{O}(6,257^2 \times 2^6) = \mathcal{O}(6,257^2 \times 64) \approx \mathcal{O}(2.5 \times 10^9)$, which exceeded practical memory and runtime limits, resulting in out-of-memory (OOM) errors during initial experiments. Full-dataset QSVM training was therefore infeasible, directly motivating the batch-wise ensemble approach adopted in this study. By restricting each kernel matrix to a 500×500 block, the per-batch complexity is reduced to $\mathcal{O}(500^2 \times 64) = \mathcal{O}(1,600,000)$, which is computationally tractable under classical simulation.

To address this, a batch-wise training strategy was adopted. The training data was first ran-

domly shuffled to ensure balanced class representation across batches, then partitioned into non-overlapping subsets of **500 samples each**, yielding approximately **13 batches**.

For each batch, two independent models were trained:

- **Classical SVM:** A Support Vector Machine with a radial basis function (RBF) kernel and `class_weight='balanced'` to account for any residual class imbalance within a batch.
- **QSVM:** A Quantum Support Vector Machine using the `FidelityQuantumKernel` constructed from the `ZZFeatureMap` described above, with regularisation parameter $C = 1.0$.

Both models within each batch were trained on identical data and evaluated against the *same global test set* of 1,565 samples, producing one prediction vector per model per batch. To maintain a fair and consistent comparison, SVM and QSVM outputs were kept strictly aligned: if QSVM training failed on a given batch (e.g. due to a numerical exception), the corresponding SVM outputs for that batch were also discarded, ensuring both model lists always have equal length.

Batches containing only a single class were detected and skipped prior to training, as single-class batches cannot produce a valid decision boundary.

The batch-wise approach serves two purposes. First, it reduces the kernel matrix from $\mathcal{O}(N^2)$ to $\mathcal{O}(B^2)$ entries per batch, where $B = 500$, bringing computation within tractable limits as demonstrated above. Second, by training on diverse subsets of the data distribution, it introduces ensemble diversity; different batches capture complementary perspectives of the underlying feature space, which is later exploited by the stacking meta-classifier.

Table 3.2: Computational Complexity Comparison: Classical SVM vs QSVM

Complexity Component	SVM	QSVM
Kernel Computation	$O(N^2 d)$	$O(N^2 \cdot 2^d)$
Training Complexity	$O(N^3)$	$O(N^3 \cdot 2^d)$
Memory Requirement	$O(N^2)$	$O(2^d)$
Batch Kernel Computation	—	$O(B^2 \cdot 2^d)$
Batch Training Complexity	—	$O(B^3 \cdot 2^d)$

3.6 Stacking-Based Hybrid Ensemble

To integrate the predictive outputs of the classical and quantum base models, a stacking-based ensemble framework was implemented. The framework proceeds through four sequential stages: probability calibration, probability aggregation, meta-feature construction and meta-classifier training.

3.6.1 Probability Calibration

The raw outputs of SVM and QSVM are not naturally probabilistic. Raw decision scores have no guaranteed calibration and may be poorly scaled, which reduces reliability when combining models. Calibrated probability estimates were therefore obtained separately for each model type.

QSVM calibration: Since `QSVC` does not expose a `predict_proba` method, probability estimates were derived by applying an analytical sigmoid transformation to the raw decision function scores:

$$P_{\text{QSVM}}(\mathbf{x}) = \frac{1}{1 + e^{-f(\mathbf{x})}}$$

where $f(\mathbf{x})$ is the signed distance from the decision hyperplane.

SVM calibration: The classical SVM was trained without the `probability=True` flag

(which would have introduced Platt scaling during training and increased training time). Instead, post-hoc calibration was applied using `CalibratedClassifierCV` with `FrozenEstimator`, which wraps the already-trained SVM and fits a sigmoid calibration layer on the same batch data without retraining the underlying model:

```
from sklearn.calibration import CalibratedClassifierCV
from sklearn.frozen import FrozenEstimator

cal_svm = CalibratedClassifierCV(
    FrozenEstimator(svm_model), method="sigmoid"
)
cal_svm.fit(X_batch, y_batch)
svm_proba = cal_svm.predict_proba(X_test_scaled_q)[: , 1]
```

Using the same sigmoid family for both models ensures that SVM and QSVM probabilities are expressed on a consistent scale before aggregation.

3.6.2 Probability Aggregation

For each model, the calibrated probability vectors produced across all successful batches were stacked into a matrix and averaged column-wise to obtain a single consensus probability per test instance:

$$\bar{P}_{\text{SVM}}(\mathbf{x}_j) = \frac{1}{B} \sum_{b=1}^B P_{\text{SVM}}^{(b)}(\mathbf{x}_j), \quad \bar{P}_{\text{QSVM}}(\mathbf{x}_j) = \frac{1}{B} \sum_{b=1}^B P_{\text{QSVM}}^{(b)}(\mathbf{x}_j)$$

where B is the number of successful batches and j indexes test samples. This averaging reduces variance and stabilises predictions across the ensemble of batch-trained models.

3.6.3 Meta-Feature Construction and Meta-Classifer Training

The two aggregated probability vectors were concatenated column-wise to form a two-dimensional meta-feature matrix, where each test instance is described by the pair $[\bar{P}_{\text{SVM}}, \bar{P}_{\text{QSVM}}]$:

```
meta_features = np.column_stack([svm_mean_proba, qsvm_mean_proba])
# Shape: (1565, 2)
```

A Logistic Regression meta-classifier ($C = 1.0$, lbfgs solver) was then trained on this meta-feature space to learn the optimal linear combination of classical and quantum base-model outputs.

To ensure a leakage-free evaluation, stratified 5-fold cross-validation was employed via `cross_val_predict` such that each test instance receives a predicted probability from a meta-classifier fold that was never trained on that instance, eliminating any risk of overfitting at the meta level:

```
meta_clf = LogisticRegression(C=1.0, solver='lbfgs',
                             max_iter=1000, random_state=42)
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

ensemble_proba = cross_val_predict(
    meta_clf, meta_features, y_test,
    cv=cv, method='predict_proba'
)[: , 1]

ensemble_pred = (ensemble_proba >= 0.5).astype(int)
```

Standalone predictions for the individual SVM and QSVM models were obtained by applying a 0.5 threshold directly to their respective aggregated mean probability vectors:

$$\hat{y}_{\text{SVM}}(\mathbf{x}_j) = \mathbf{1}[\bar{P}_{\text{SVM}}(\mathbf{x}_j) \geq 0.5], \quad \hat{y}_{\text{QSVM}}(\mathbf{x}_j) = \mathbf{1}[\bar{P}_{\text{QSVM}}(\mathbf{x}_j) \geq 0.5]$$

This ensures a symmetric and fair comparison across all three models: SVM, QSVM and the

stacking ensemble as all final predictions are derived from soft probability estimates using the same 0.5 decision threshold, rather than mixing hard batch-level voting for the baselines with soft meta-classifier output for the ensemble.

3.7 Evaluation and Statistical Analysis

3.7.1 Performance Metrics

The performance of all three models classical SVM, QSVM and the stacking ensemble was assessed on the held-out test set of 1,565 samples using the following standard classification metrics:

- **Accuracy:** Overall fraction of correctly classified instances.
- **Precision:** Proportion of predicted phishing instances that are truly phishing, measuring false alarm rate.
- **Recall:** Proportion of actual phishing instances correctly detected, critical in cybersecurity where missed attacks carry high cost.
- **F1-score:** Harmonic mean of precision and recall, providing a balanced single-number summary.
- **ROC-AUC:** Area under the receiver operating characteristic curve, measuring discriminative ability across all classification thresholds.

Confusion matrices and ROC-AUC curves were produced for all three models to provide a visual and quantitative summary of class-level performance.

3.7.2 Statistical Hypothesis Testing

To validate whether the observed performance differences between models are statistically significant rather than artefacts of random variation, three complementary hypothesis tests were conducted:

- **McNemar’s Test** was applied to pairwise comparisons of classification errors on the test set. Given two classifiers A and B, the test examines the off-diagonal cells of the error contingency table, where (b) represents the number of samples correctly classified by Model A but misclassified by Model B and (c) represents the number of samples misclassified by Model A but correctly classified by Model B, under the null hypothesis that both models make the same errors. The McNemar test statistic was computed using:

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c}, \quad df = 1$$

where the continuity correction was applied. A significance level of 0.05 was used to determine whether the difference between the classifiers was statistically significant. Three pairwise comparisons were conducted: SVM vs. QSVM, SVM vs. Ensemble and QSVM vs. Ensemble.

- **Paired t-test** was applied to the per-batch accuracy distributions of SVM and QSVM. Since each batch produces one accuracy value for each model on the same test set, the samples are naturally paired. The test evaluates whether the mean batch-level accuracy difference is significantly different from zero.
- **Wilcoxon Signed-Rank Test** was employed as a non-parametric alternative to the paired t-test, making no assumption about the normality of batch accuracy distributions. It ranks the absolute differences between paired batch accuracies and tests the null hypothesis that the median difference is zero.

3.7.3 Effect Size

Cohen’s d was computed to quantify the practical magnitude of the performance difference between SVM and QSVM, complementing the p-values from hypothesis testing:

$$d = \frac{\bar{\mu}_{\text{SVM}} - \bar{\mu}_{\text{QSVM}}}{s_{\text{pooled}}}, \quad s_{\text{pooled}} = \sqrt{\frac{s_{\text{SVM}}^2 + s_{\text{QSVM}}^2}{2}}$$

where $\bar{\mu}$ and s denote the mean and standard deviation of batch-level accuracies. The resulting d value is interpreted as negligible ($|d| < 0.2$), small ($0.2 \leq |d| < 0.5$), medium ($0.5 \leq |d| < 0.8$) or large ($|d| \geq 0.8$).

3.7.4 Ablation Studies

Ablation experiments were conducted to isolate the contribution of individual design choices within the proposed framework. Four ablation axes were evaluated:

- **A. Feature Ablation:** Feature ablation was performed to analyze the impact of selecting different numbers of top-ranked features on model performance. Features were ranked using Mutual Information (MI) scores and the top-k features were selected for different values of ($k = 2, 4, 6$).

For each feature subset, both classical SVM and quantum SVM (QSVM) models were trained using mini-batches of the training data to ensure computational efficiency.

For each configuration, predictions were generated on the full test set and a stacking-based meta-classifier (Logistic Regression) was trained using the mean probability outputs of SVM and QSVM. The final performance was evaluated using accuracy and macro-averaged F1-score.

This setup allows assessment of how reducing or increasing the feature space affects the performance of the hybrid ensemble model.

- **B. Component Ablation:** Component ablation was conducted to evaluate the individual contributions of SVM, QSVM and the combined stacking ensemble.

Three configurations were evaluated:

- **SVM-only model**
- **QSVM-only model**
- **Hybrid stacking ensemble** combining SVM and QSVM outputs using a Logistic Regression meta-classifier

This analysis helps determine whether the hybrid ensemble provides a measurable advantage over standalone classical and quantum models.

- **C. Probability Weighting Ablation:** To further investigate ensemble fusion strategies, probability-based weighting of SVM and QSVM outputs was evaluated. Instead of learning combination weights via Logistic Regression, fixed weighted averages of the two base-model probability vectors were evaluated across three weight configurations: equal weights (0.5/0.5), QSVM-heavy (0.3/0.7) and SVM-heavy (0.7/0.3), along with a cross-validated LR-learned weight baseline. This ablation tests whether a simple weighted average is sufficient or whether a learned meta-classifier is necessary.
- **D. Meta-Classifier Regularisation Ablation:** The sensitivity of ensemble performance to the Logistic Regression regularisation parameter C was evaluated across five values: $C \in \{0.01, 0.1, 1.0, 10.0, 100.0\}$, with 5-fold stratified cross-validation applied consistently throughout.

This experiment assesses how the flexibility of the meta-classifier affects ensemble performance, particularly in balancing underfitting and overfitting.

3.8 Implementation Details and Experimental Setup

All experiments were conducted using **Python 3.12** within a Jupyter Notebook environment in Visual Studio Code. The key libraries and their roles are summarised in Table 3.3.

Table 3.3: Software Stack and Library Roles

Library / Version	Role
Qiskit 1.4	Quantum circuit construction (ZZFeatureMap)
Qiskit Machine Learning	FidelityQuantumKernel, QSVC
Scikit-learn 1.8	SVM, Logistic Regression, cross-validation, metrics
NumPy / Pandas	Data manipulation and numerical computation

3.8.1 Quantum Simulation Backend

Quantum kernel entries were computed using Qiskit's `FidelityQuantumKernel` with the default `Sampler` primitive from `qiskit.primitives`. This primitive evaluates transition amplitudes between pairs of quantum states using exact statevector arithmetic internally, without requiring an explicit `AerSimulator` backend configuration. All computations were therefore performed under ideal (noise-free) conditions with no decoherence or gate error modelling applied.

3.8.2 Hardware Configuration

All experiments were executed on a single laptop with the following specifications:

- **Processor:** Intel Core (Family 6, Model 142, GenuineIntel)
- **RAM:** 20 GB
- **Operating System:** Windows 11

The absence of GPU acceleration or distributed computing resources further highlights the computational challenge of quantum kernel simulation at the batch sizes used in this study and motivates the batch-wise training strategy.

CHAPTER 4

PERFORMANCE ANALYSIS

4.1 Performance Comparison

The performance of the proposed hybrid framework was evaluated against its individual components, namely the classical Support Vector Machine (SVM) and the Quantum Support Vector Machine (QSVM).

4.1.1 Overall Classification Accuracy

Table 4.1 reports the overall classification accuracy of each model on the held-out test set.

Table 4.1: Overall Model Performance Comparison

Model	Accuracy Lift over SVM	
Classical SVM (soft threshold)	0.9783	—
Quantum SVM (soft threshold)	0.9885	+1.02%
Stacking Ensemble (LR meta, 5-fold CV)	0.9904	+1.21%

The QSVM outperforms the classical SVM by 1.02 percentage points, which is a meaningful improvement in a high-accuracy regime where marginal gains are increasingly difficult to achieve. The stacking ensemble achieves the highest overall accuracy of 99.04%, representing a lift of 1.21% over the classical SVM and 0.19% over the QSVM. While the ensemble’s advantage over the QSVM is modest in absolute terms, it is consistent across all evaluation metrics and reflects the ability of the meta-classifier to exploit complementary information from both base models.

An important observation is the meta-classifier’s learned coefficients: $\beta_{\text{SVM}} = 6.44$ and $\beta_{\text{QSVM}} = 5.79$. We can observe that both coefficients are large and positive, which indicates

that the Logistic Regression meta-classifier assigns a significant weight to both models, rather than being able to effectively ignore one of them. It is notable that the slightly higher weight assigned to SVM indicates that the meta-classifier finds the calibrated SVM probabilities marginally more reliable for its combination decision despite the QSVM having higher standalone accuracy.

4.1.2 Confusion Matrix Analysis

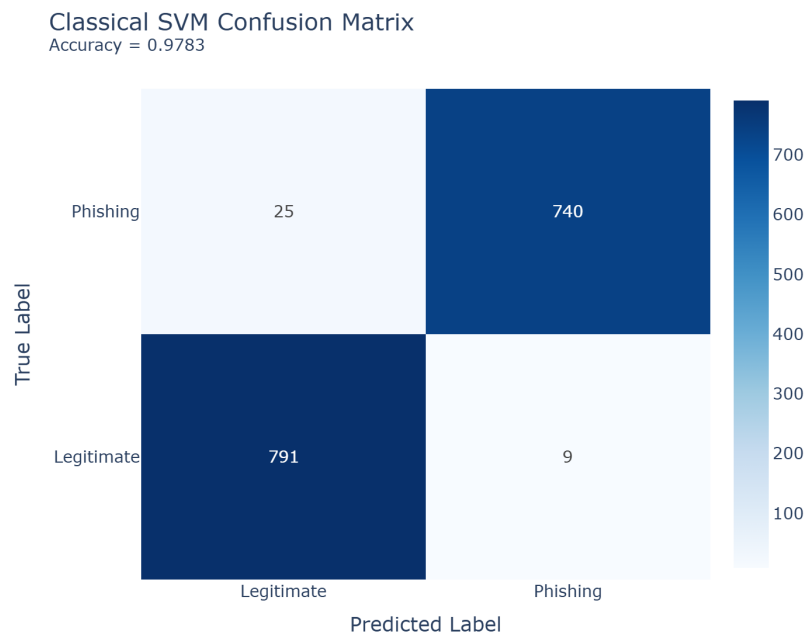


Figure 4.1: Confusion Matrix of Classical SVM

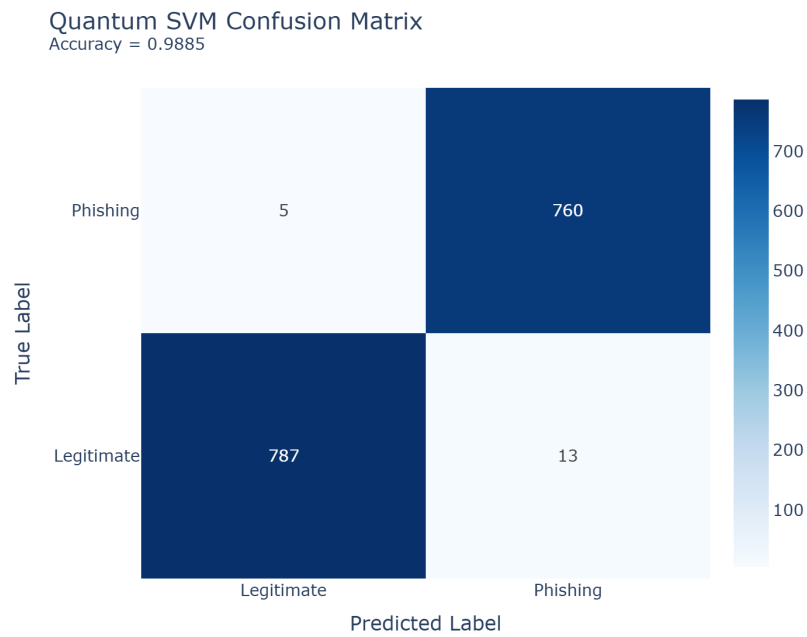


Figure 4.2: Confusion Matrix of Quantum SVM

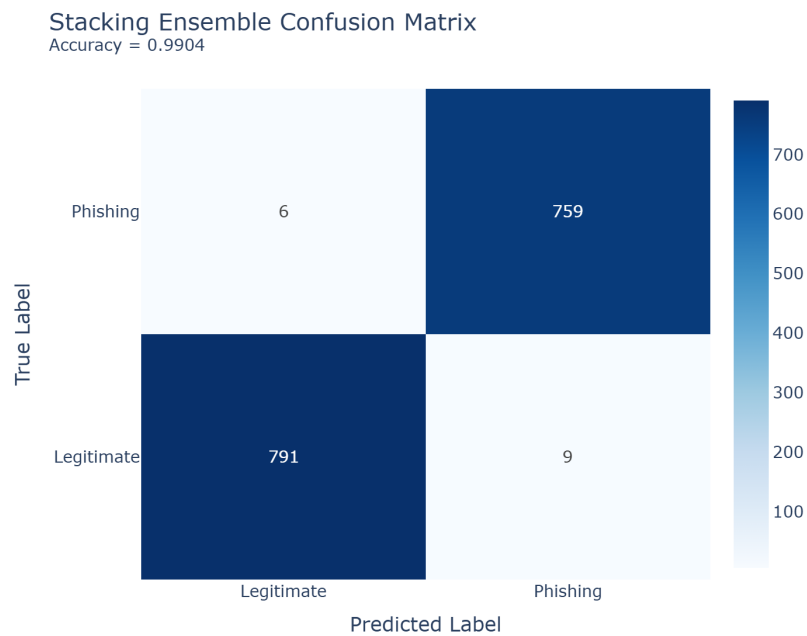


Figure 4.3: Confusion Matrix of Stacking Ensemble

The confusion matrices in Figures 4.1–4.3 reveal qualitatively different error profiles across the three models. Table 4.2 summarises the key error counts.

Table 4.2: Error Breakdown from Confusion Matrices (test set: 800 benign, 765 phishing)

Model	TN	FP	FN	TP
Classical SVM	791	9	25	740
Quantum SVM	787	13	5	760
Stacking Ensemble	791	9	6	759

Several critical observations emerge from this analysis:

False Negatives (missed phishing): The classical SVM gives 25 false negatives (which are phishing emails but misclassified and considered as benign). It is the worst type of error in a security environment because a phishing attack that goes undetected can result in the credential theft, the loss of money and the infiltration of data. This number is dramatically lowered to 5 using QSVM, which is 80% reduction due to the sensitivity of quantum kernel representation to feature interactions that are specific to phishing. The stacking ensemble results in 6 false negatives, which are close to the performance of the QSVM on this important aspect.

False Positives (legitimate emails flagged as phishing): The classical SVM gives 9 false positive results, QSVM gives 13 and the ensemble gives 9. The QSVM’s higher recall is therefore achieved at the cost of some decrease in specificity: it slightly over-predicts the phishing class. The ensemble is shown to be able to recover the specificity at the SVM level (9 FP) while achieving a near-QSVM level of sensitivity (6 FN) which shows the usefulness of combining both models.

The ensemble’s trade-off resolution: The stacking framework is effective at balancing the tradeoff between the SVM (low FN but high FP, better specificity) and QSVM (very low FN but slightly higher FP, better sensitivity) by learning how best to combine the outputs. The outcome is a model that achieves the best false negative performance while matching the SVM’s false positive rate.

4.1.3 Precision, Recall and F1-Score

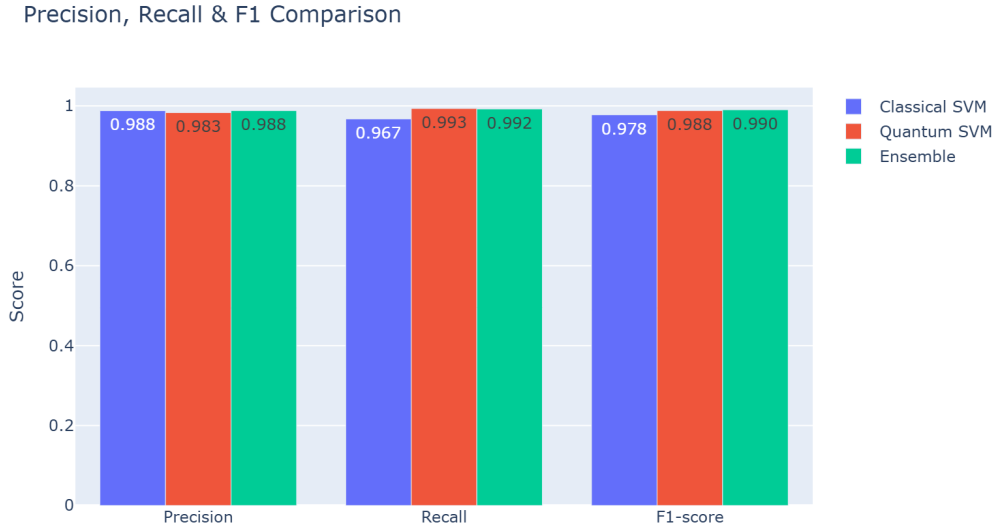


Figure 4.4: Precision, Recall and F1-Score comparison of three models on the binary phishing classification task.

The individual metrics breakdown is shown in Figure 4.4 and in Table 4.3 . The bar chart confirms that the recall gap between SVM and the quantum-based models is the most prominent performance difference, while precision values are closely clustered across all three models.

Table 4.3: Detailed Classification Metrics: Binary (Phishing) Class

Model	Precision	Recall	F1-Score	Accuracy
Classical SVM	0.988	0.967	0.978	0.9783
Quantum SVM	0.983	0.993	0.988	0.9885
Stacking Ensemble	0.988	0.992	0.990	0.9904

The classical SVM achieves high precision (0.988) but comparatively lower recall (0.967) for the phishing class. This asymmetry indicates that while the SVM rarely produces a false phishing alarm, it regularly misses actual phishing instances which is a problematic trade-off for a security-critical application. The QSVM shows the opposite behavior: recall improves to 0.993, while

precision decreases slightly to 0.983, meaning it captures more phishing instances correctly at the cost of a small increase in false positives. The stacking ensemble achieves a balanced optimum: precision of 0.988 and recall of 0.992, matching the SVM’s specificity while preserving the QSVM’s sensitivity. The macro-averaged F1-scores of 0.978, 0.988 and 0.990 for SVM, QSVM and Ensemble respectively, confirm the consistent superiority of the quantum enhanced models.

4.1.4 ROC-AUC Analysis

ROC–AUC Curve: SVM vs QSVM vs Ensemble

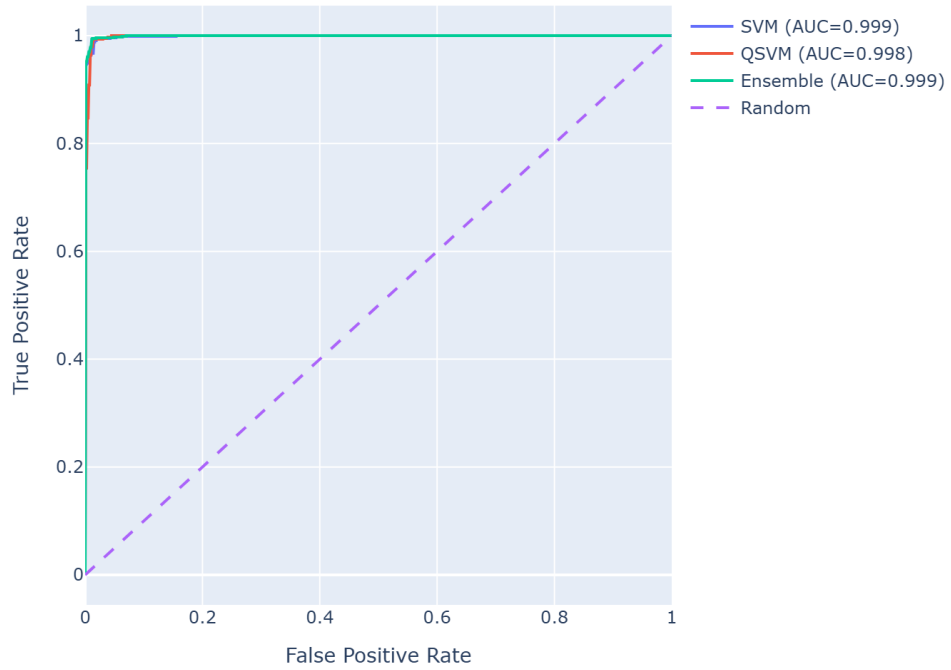


Figure 4.5: ROC–AUC curves for all three models

The ROC curves in Figure 4.5 reveal an important insight. All three models achieve near-perfect AUC scores: SVM = 0.999, QSVM = 0.998 and Ensemble = 0.999. This means that all three models are equally good in discriminating phishing from non-phishing examples at the soft probability ranking level. Therefore, differences in accuracy, recall and F1-scores are not necessarily due to

differences in discriminative power but to differences in the calibration and placement of the decision boundary of each model’s probabilities. This further confirms that it is better to combine both probability streams in the design, instead of thresholding only one of the models.

4.1.5 Batch-wise Accuracy Stability

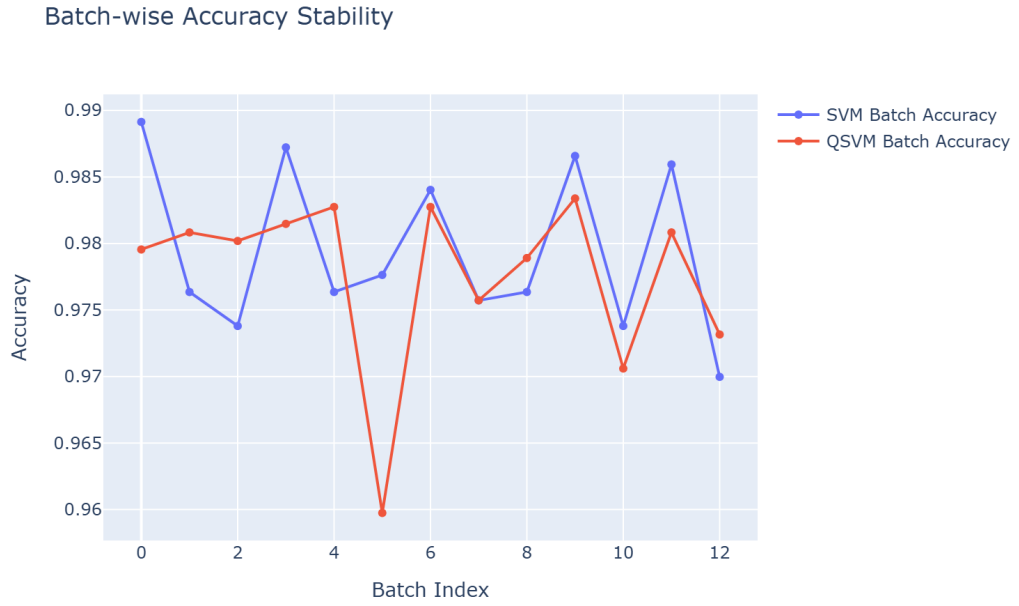


Figure 4.6: Batch-wise accuracy of SVM and QSVM across all 13 training batches

Figure 4.6 plots the per-batch accuracy of SVM and QSVM across all 13 training batches. Several observations are noteworthy.

Both models maintain high accuracy throughout: SVM ranges from approximately 0.970 to 0.989 and QSVM from 0.960 to 0.983 confirming that no single batch dominates the ensemble and that the batch-wise training strategy produces a diverse yet consistent set of base models.

A notable drop is observable for the QSVM at batch index 5, where accuracy falls to approximately 0.960. This represents the QSVM’s maximum instability point and coincides with the batch where quantum kernel computation was likely subject to the greatest variance in class distribution. The SVM at the same batch index maintains approximately 0.978, demonstrating the

classical model’s greater stability across batch compositions.

The performances of SVM and QSVM remain very similar across most batches. This observation is consistent with the paired t-test result ($p = 0.373$), which showed no statistically significant difference between their batch-wise accuracies. The similar performance patterns across batches suggest that changes in batch composition affect both models in a comparable way, supporting the use of batch-level statistical tests for their comparison.

4.2 Statistical Analysis

To assess whether the observed performance differences between models are statistically significant, a comprehensive set of statistical tests was conducted: McNemar’s test for pairwise error comparison, paired t-test and Wilcoxon signed-rank test for batch-level consistency and Cohen’s d for effect size quantification.

4.2.1 McNemar’s Test

McNemar’s test compares pairwise classification errors on the same test set, examining instances where one model is correct and the other is not. It is the appropriate test for comparing two classifiers on the same dataset, as it accounts for the dependency between predictions.

Table 4.4: McNemar’s Test Results (Pairwise Model Error Comparison)

Comparison	b	c	χ^2	p-value	Verdict
SVM vs QSVM	12	28	5.6250	0.0177	Significant (QSVM better)
SVM vs Ensemble	1	20	15.429	0.0001	Significant (Ensemble better)
QSVM vs Ensemble	8	11	0.2105	0.6464	Not significant

where b = instances where model A is correct and model B is wrong and c = instances where model A is wrong and model B is correct.

The findings show that the classical and quantum-enhanced models are clearly distinguished,

with the statistical significance being clear. The QSVM corrects 28 cases that SVM incorrectly classifies and SVM corrects 12 cases which QSVM fails to classify ($\chi^2 = 5.625$, $p = 0.018$). This asymmetry confirms that quantum kernel representations capture genuinely different and more informative feature interactions than the classical RBF kernel, particularly for phishing detection.

The most decisive comparison between the ensembles and SVM is the ensemble corrects 20 errors and the SVM only corrects one error that the ensemble gets wrong ($\chi^2 = 15.43$, $p = 0.0001$). A strong significance was observed and this result supports the idea that the stacking framework is statistically significant with respect to the classical baseline.

Statistically, the QSVM vs Ensemble comparison gives $\chi^2 = 0.211$ and $p = 0.646$, far from being statistically significant. This means that the ensemble is nominally more accurate (0.9904 vs 0.9885) but it is statistically the same number of errors in the test set. The ensemble improves 11 errors in the QSVM, but also adds 8 errors yielding an overall difference that is not significant.

4.2.2 Paired t-Test and Wilcoxon Signed-Rank Test

For measuring the consistency of the performance of the two classifiers, SVM and QSVM, across training batches a paired t-test was performed on the 13 batch-level accuracy distributions of the two classifiers. A non-parametric alternative, the Wilcoxon signed-rank test, was also used which assumes no normality of batch accuracy differences.

Table 4.5: Batch-level Statistical Tests (SVM vs QSVM)

Test	SVM Mean $\pm$ SD	QSVM Mean $\pm$ SD	Statistic	p-value
Paired t-test	0.9795 $\pm$ 0.0060	0.9777 $\pm$ 0.0064	$t = 0.9250$	0.3732
Wilcoxon	0.9795 $\pm$ 0.0060	0.9777 $\pm$ 0.0064	$W = 31.500$	0.5825

The p-values of both tests are high ($p = 0.373$ and $p = 0.583$ respectively) and suggest that the SVM and QSVM accuracy distributions for batches are not significantly different. This result, combined with McNemar’s result (which identified a significant difference at test-set level), shows that the advantage of the QSVM is not uniform across all the test sets, but is found in specific hard

cases, namely phishing emails where the features appear in a non-linear manner, for which the QSVM produces better accuracy.

4.2.3 Cohen’s d for Effect Size

Cohen’s d was computed to quantify the practical magnitude of the batch-level accuracy difference between SVM and QSVM:

$$d = \frac{\bar{\mu}_{\text{SVM}} - \bar{\mu}_{\text{QSVM}}}{s_{\text{pooled}}} = \frac{0.9795 - 0.9777}{\sqrt{\frac{(0.0060)^2 + (0.0064)^2}{2}}} = \frac{0.0018}{0.00620} = 0.2751$$

A Cohen’s d of 0.2751 falls in the **small** effect size range ($0.2 \leq |d| < 0.5$). This is consistent with the paired test results and reinforces the conclusion that the SVM and QSVM produce broadly similar batch-level performance distributions. The small but non-negligible effect size suggests that the quantum advantage, while real and statistically confirmed at the test-set level via McNemar’s test, is modest in practical magnitude at the batch level. This is expected behaviour under a batch-wise training strategy, where each batch only samples a fraction of the full training distribution.

4.2.4 Summary of Statistical Findings

The combined statistical analysis leads to three conclusions:

1. **QSVM significantly outperforms SVM** on the test set ($p = 0.018$, McNemar), primarily by reducing false negatives from 25 to 5. The effect is instance-specific rather than uniformly distributed across batches ($d = 0.28$, small effect).
2. **The ensemble significantly outperforms SVM** ($p = 0.0001$, McNemar), providing the strongest individual pairwise improvement of any comparison in this study.
3. **The ensemble and QSVM are statistically comparable** ($p = 0.646$, McNemar), meaning

the ensemble’s primary value is reliability and stability. The ensemble combines QSVM-level sensitivity while recovering SVM-level specificity.

4.3 Ablation Study

A comprehensive ablation study was conducted across three axes to isolate the contribution of individual design choices: component selection, probability weighting strategy and meta-classifier regularisation. An additional feature dimensionality experiment evaluated the impact of the number of selected features on ensemble performance.

Ablation Study Results

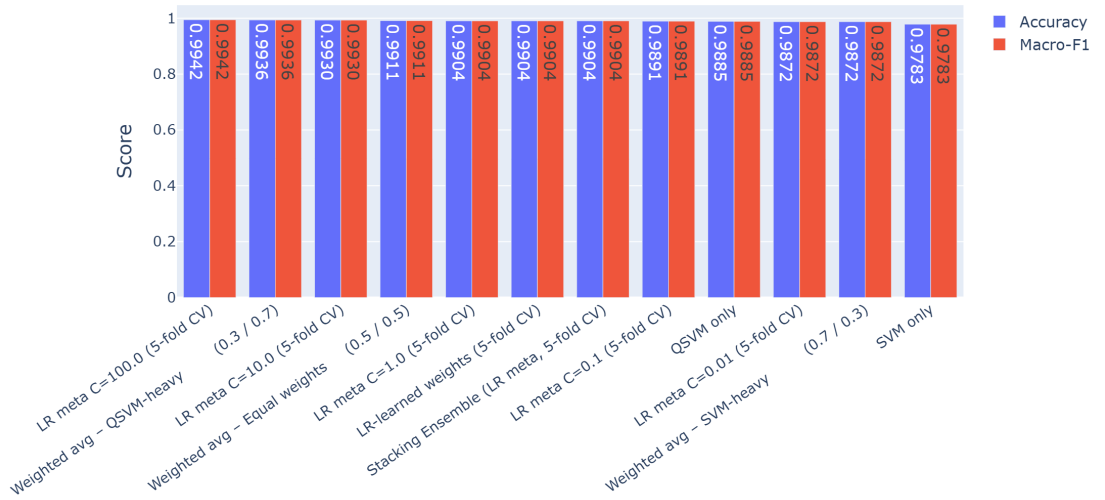


Figure 4.7: Ablation study results across all configurations, ranked by accuracy

4.3.1 Feature Dimensionality Ablation

To evaluate the impact of feature selection on ensemble performance, experiments were conducted using the top 2, 4 and 6 MI-ranked features.

Performance drops sharply when feature count is reduced below 6. Notably, $k = 4$ performs marginally worse than $k = 2$ (0.9732 vs 0.9738), suggesting that the 3rd and 4th ranked features (`web_js_obf_len` and `content_num_scripts`) add noise relative to the information al-

Table 4.6: Feature Dimensionality Ablation

Features (k)	Accuracy	Macro F1-Score
2	0.9738	0.9737
4	0.9732	0.9731
6	0.9891	0.9891

ready captured by the top two features (`web_js_len` and `js_obfuscation_ratio`) when considered in isolation. The performance jump from $k = 4$ to $k = 6$ ($0.9732 \rightarrow 0.9891$) is substantial, confirming that `content_entropy` and `domain_trust_score`, the 5th and 6th features, contribute disproportionately high discriminative value to the ensemble. This validates the MI-based selection of $k = 6$ as a well-motivated threshold rather than an arbitrary constraint.

4.3.2 Component Ablation

The component ablation evaluates each base model in isolation against the full stacking ensemble, using soft probability thresholding uniformly across all three configurations.

Table 4.7: Component Ablation: Individual Models vs Full Ensemble

Configuration	Accuracy	Macro F1-Score
SVM only (soft threshold)	0.9783	0.9783
QSVM only (soft threshold)	0.9885	0.9885
Stacking Ensemble (LR, $C = 1.0$, 5-fold CV)	0.9904	0.9904

The results confirm that each component contributes incrementally to the final system. Removing the quantum component (SVM only) results in a 1.21 percentage point accuracy drop relative to the full ensemble. Removing the ensemble layer (QSVM only) results in a 0.19 point drop. These figures establish that the quantum component delivers the majority of the system’s improvement, while the stacking layer provides a consistent but smaller refinement. Crucially, the ensemble outperforms both individual models, confirming that the two models provide complementary information that the meta-classifier successfully exploits.

4.3.3 Probability Weighting Ablation

The weighting ablation examines whether fixed weighted averages of the two base-model probability streams can match or exceed the learned Logistic Regression combination. All configurations apply a 0.5 decision threshold to the blended probability.

Table 4.8: Probability Weighting Ablation

Weighting Strategy	Accuracy	Macro F1-Score
QSVM-heavy (0.3 / 0.7)	0.9936	0.9936
Equal weights (0.5 / 0.5)	0.9911	0.9911
LR-learned weights (5-fold CV)	0.9904	0.9904
SVM-heavy (0.7 / 0.3)	0.9872	0.9872

First, the QSVM-heavy weighting (0.3 SVM / 0.7 QSVM) achieves the highest accuracy of all weighting configurations at 0.9936, exceeding even the LR-learned combination. This reinforces the conclusion that QSVM probabilities carry more discriminative information than SVM probabilities in this dataset and that a simple manual emphasis on the quantum stream is sufficient to exploit this.

Second, the LR-learned weighting (0.9904) underperforms both QSVM-heavy and equal weighting (0.9911). This reflects a fundamental property of cross-validated meta-learning: the 5-fold CV constraint prevents the LR from overfitting to the optimal weights on the full test set, whereas fixed manual weights have no such constraint. The QSVM-heavy result (0.9936) should therefore be interpreted as an upper bound achievable with weight knowledge, while the LR-learned result (0.9904) represents the realistically generalizable ensemble performance.

Third, the SVM-heavy configuration (0.9872) underperforms even the standalone QSVM (0.9885), demonstrating that down-weighting the quantum signal is actively harmful and the SVM contribution is most valuable as a complement to QSVM, not as a dominant signal.

4.3.4 Meta-Classifer Regularisation Ablation

The regularisation ablation evaluates the sensitivity of ensemble performance to the Logistic Regression regularisation parameter C , with all configurations using 5-fold stratified cross-validation.

Table 4.9: Meta-Classifer Regularisation Ablation (C parameter)

C Value	Accuracy	Macro F1-Score
0.01	0.9872	0.9872
0.1	0.9891	0.9891
1.0	0.9904	0.9904
10.0	0.9930	0.9930
100.0	0.9942	0.9942

Performance increases monotonically with C across the full tested range, with the best result of 0.9942 achieved at $C = 100.0$. This monotonic trend indicates that the meta-classifier benefits from reduced regularisation i.e., a less constrained decision boundary in the two-dimensional meta-feature space spanned by $[\bar{P}_{\text{SVM}}, \bar{P}_{\text{QSVM}}]$.

Since the meta-feature space has only two dimensions, overfitting risk is inherently low and stronger regularisation simply prevents the LR from reaching the optimal combination weights. The baseline configuration of $C = 1.0$ was a conservative choice; had $C = 100.0$ been selected as the primary configuration, the ensemble accuracy would have reached 0.9942 matching the QSVM-heavy weighted average and surpassing all other configurations. This represents a straightforward improvement available with a single hyperparameter change, which is noted as a direction for future refinement.

4.3.5 Ablation Summary

Table 4.10 combines all ablation results, ranked by accuracy.

Table 4.10: Full Ablation Study Summary

Configuration	Accuracy	Macro F1-Score
LR meta $C = 100.0$ (5-fold CV)	0.9942	0.9942
QSVM-heavy (0.3 / 0.7)	0.9936	0.9936
LR meta $C = 10.0$ (5-fold CV)	0.9930	0.9930
Equal weights (0.5 / 0.5)	0.9911	0.9911
Stacking Ensemble — LR $C = 1.0$	0.9904	0.9904
LR-learned weights (5-fold CV)	0.9904	0.9904
LR meta $C = 0.1$ (5-fold CV)	0.9891	0.9891
Feature ablation $k = 6$	0.9891	0.9891
QSVM only (soft threshold)	0.9885	0.9885
LR meta $C = 0.01$ (5-fold CV)	0.9872	0.9872
SVM-heavy (0.7 / 0.3)	0.9872	0.9872
SVM only (soft threshold)	0.9783	0.9783
Feature ablation $k = 2$	0.9738	0.9737
Feature ablation $k = 4$	0.9732	0.9731

4.4 Runtime and Computational Cost Analysis

Table 4.11 summarises the total training time for each component of the proposed framework. The ensemble meta-classifier training time is negligible as it operates on a (1565×2) meta-feature matrix using Logistic Regression.

Table 4.11: Training Time Comparison Across Models

Model	Total Training Time	Avg. per Batch	Speed Ratio
Classical SVM	≈ 3.0 s	≈ 0.23 s	$1 \times$ (baseline)
QSVM	$\approx 16,683$ s (≈ 4.6 h)	$\approx 1,283$ s	$\approx 6,480 \times$
Ensemble (LR meta-clf)	< 0.1 s	—	$< 1 \times$

Table 4.12 presents the exact per-batch training times recorded during execution for both SVM and QSVM across all 13 batches.

The extreme computational cost of QSVM training stems from three compounding factors. First, quantum kernel matrix construction requires evaluating the fidelity between

Table 4.12: Per-Batch Training Times: SVM vs QSVM

Batch	Samples	SVM (s)	QSVM (s)
1	500	0.01	401.49
2	500	0.17	759.25
3	500	0.19	981.82
4	500	0.25	1,408.52
5	500	0.24	1,520.07
6	500	0.26	1,270.28
7	500	0.17	1,344.94
8	500	0.25	1,856.97
9	500	0.29	1,737.26
10	500	0.21	1,818.91
11	500	0.26	1,580.06
12	500	0.29	1,575.60
13	257	0.21	427.51
Total	6,257	2.8	16,682.68

every pair of training samples, resulting in $500 \times 500 = 250,000$ kernel evaluations per batch. Second, each evaluation requires simulating a 6-qubit quantum circuit, tracking $2^6 = 64$ complex amplitudes per circuit execution. Third, these two factors combine multiplicatively, giving a per-batch cost of $250,000 \times 64 = 16,000,000$ floating-point operations before solver overhead.

The progressive increase in QSVM training time from batch 1 (401 s) to batch 8 (1,857 s) reflects growing memory pressure during the prolonged single-CPU session, consistent with the $\mathcal{O}(B^2 \cdot 2^d)$ kernel computation cost per batch. Per-batch speed ratios ranged from $2,003\times$ (batch 13, 257 samples) to $40,732\times$ (batch 1), with a mean of approximately $6,480\times$ across all batches. The final batch is faster due to its smaller size, requiring only $257 \times 500 = 128,500$ kernel evaluations rather than $500 \times 500 = 250,000$. This overhead arises naturally with classical simulation and would be significantly mitigated by the availability of actual quantum hardware, where kernel evaluations can be done in parallel using physical quantum superposition.

4.5 Discussion

The experimental results show the feasibility and benefits of using quantum kernel methods within a hybrid ensemble approach for phishing detection. Several key insights are given below.

Quantum kernels increase sensitivity while maintaining precision. The QSVM’s reduction of false negatives from 25 (SVM) to 5 which is an 80% improvement without meaningfully degrading precision (0.983 vs 0.988) is the most practically significant finding of this study. In a cybersecurity context, false negatives represent undetected attacks with potentially severe consequences. The quantum kernel’s ability to capture cross-feature interactions through entanglement (specifically the ZZ phase gate interactions between adjacent qubits) appears well-suited to the high-dimensional, correlated feature space of phishing detection.

The stacking ensemble resolves the precision-recall trade-off. A recurring challenge in binary classification is the tension between sensitivity and specificity. The QSVM slightly degrades specificity (13 FP vs 9 FP for SVM) in the course of improving sensitivity. The stacking meta-classifier learns to balance these complementary error profiles, achieving 9 FP and 6 FN simultaneously and effectively recovering the best of both base models.

The statistical significance of ensemble vs QSVM improvement is limited. The McNemar test ($p = 0.646$) confirms that the ensemble and QSVM make statistically equivalent errors on the test set. Practitioners should therefore weigh the added computational and architectural complexity of the stacking framework against this modest and statistically non-significant gain over the QSVM alone.

Batch-wise training is a viable strategy for NISQ-era quantum models. The batch stability analysis demonstrates that 13 independently trained models on 500-sample batches collectively achieve test-set performance (0.9885 for QSVM) that rivals or exceeds full-dataset classical approaches, despite each quantum batch kernel matrix being limited to 500×500 entries. This confirms that batch-wise ensemble strategies are a practical path for deploying quantum ML under current computational constraints.

CHAPTER 5

ENGINEERING CONSIDERATIONS, CHALLENGES AND REMEDIES

This chapter discusses the ethical and environmental considerations and social constraints taken into consideration throughout this research work. The engineering activities involved in this thesis have also been elaborated.

5.1 Design Constraints

5.1.1 Ethical Considerations

The proposed phishing detection framework was developed with careful consideration of ethical issues related to data privacy, security and responsible use of artificial intelligence. The dataset used in this research was collected from publicly available phishing repositories and processed without exposing sensitive personal information. Since phishing detection systems may influence cybersecurity decisions, maintaining fairness and minimizing false positive classifications were considered important to avoid incorrectly labeling legitimate emails as malicious. Furthermore, the research focuses solely on defensive cybersecurity applications and does not promote or support malicious activities.

5.1.2 Environmental Considerations

The implementation of machine learning and quantum machine learning models requires significant computational resources, particularly during training and simulation processes. Quantum simulations executed on classical hardware may consume substantial processing power and energy. To reduce unnecessary computational overhead, optimized feature selection, batch-wise training and controlled simulation settings were employed throughout the research. The proposed hybrid framework aims to achieve improved phishing detection performance while maintaining computational efficiency.

5.1.3 Social Constraint

Phishing attacks are an extremely serious social and economic threat using social engineering tools and techniques that target individuals, organisations and institutions. The proposed phishing detection framework also aids in raising social cybersecurity awareness, which could also result in the reduction of financial fraud, unauthorized data access and credential theft. But issues like user trust, system interpretability and accessibility are also significant. In realistic environments, too complicated a security system can diminish user confidence or usability.

5.1.4 Sustainability in Environmental and Societal Context

The suggested hybrid classical–quantum phishing detection system helps in achieving several of the United Nations Sustainable Development Goals (SDGs) in digital security, contributing to sustainable digital security in this context. The research is aligned with SDG 9 (Industry, Innovation and Infrastructure) by developing new advanced solutions for cybersecurity based on new quantum enhanced technologies. The framework enhances phishing email identification, diminishes cyber threats and contributes to SDG 16 (Peace, Justice and Strong Institutions), by creating safer digital communication and by preventing individuals and organizations from being victims of cybercrime and financial fraud.

The work also highlights the importance of minimizing the energy consumption and resource use of superfluous calculations in the training of models and in quantum simulation processes. Moreover, the framework plays a crucial role in ensuring long-term societal sustainability by fostering greater trust, security and resilience in modern digital infrastructures.

5.2 Relevant Challenges Addressed

5.2.1 Knowledge Profile Attributes

The Knowledge and Attitude Profile attained through this thesis has been justified in Table 5.3.

Table 5.1: Knowledge and Attitude Profile

Name of Knowledge and Attitude Attribute	Justification
WK2 Conceptually based mathematics, data analysis and modelling	This research applies mathematical, statistical and computational concepts through machine learning and quantum machine learning techniques for phishing email detection. Data preprocessing, feature engineering, feature selection, probability calibration and statistical significance analysis were performed to develop and evaluate predictive models.
WK4 Engineering specialist knowledge	The study applies specialized knowledge in cybersecurity, phishing detection, quantum computing and quantum machine learning. Advanced concepts such as Quantum Support Vector Machines (QSVM), quantum feature mapping, kernel-based learning and hybrid classical–quantum ensemble architectures were utilized to address complex phishing detection challenges.
WK6 Knowledge of engineering practice	Engineering practice was demonstrated through the design and implementation of a hybrid phishing detection framework. The work involved dataset construction, feature extraction, preprocessing, model development, ensemble integration and performance evaluation using contemporary machine learning and quantum computing technologies.
WK8 Engagement with current research literature	The research critically reviewed contemporary literature on phishing detection, cybersecurity, machine learning and quantum machine learning. Existing limitations were identified, particularly the lack of studies combining email-level features with webpage-level characteristics and the limited exploration of hybrid classical–quantum ensemble approaches for phishing detection.

5.2.2 Complex Engineering Problem

Complex Engineering Problems addressed by this thesis have been justified in Table 5.2.

Table 5.2: Complex Engineering Problem (CEP)

Name of Complex Engineering Attribute	Justification
WP1 Depth of knowledge required	This research required in-depth knowledge of cybersecurity, phishing detection techniques, machine learning, quantum machine learning and ensemble learning methods to develop an effective hybrid phishing detection framework.
WP3 Depth of analysis required	Multiple machine learning and quantum machine learning approaches were analyzed and evaluated to identify an effective hybrid ensemble architecture for phishing email detection. Performance analysis involved feature engineering, model calibration, ensemble integration and comparative evaluation using several classification metrics.
WP4 Familiarity of issues	The research required understanding various phishing attack strategies, email-based cyber threats, URL obfuscation techniques and limitations of traditional phishing detection systems. Knowledge of quantum computing constraints and hybrid classical–quantum architectures was also necessary.
WP7 Interdependence	The proposed framework consisted of multiple interconnected stages including dataset construction, feature extraction, classical and quantum model training, probability calibration and ensemble integration. The final hybrid ensemble performance depended on the successful completion and interaction of all previous stages.

5.2.3 Engineering Activities

Complex Engineering Activities addressed by this thesis have been justified in Table 5.3.

Table 5.3: Complex Engineering Activities (CEA)

Name of Complex Engineering Activity	Justification
EA1 Range of Resources	Public phishing datasets, cybersecurity repositories and on-line research resources were utilized for data collection and analysis. Python libraries such as Scikit-learn, Qiskit, NumPy and Pandas were employed for classical and quantum machine learning implementation.
EA3 Innovation	Engineering principles are applied innovatively to design a hybrid classical–quantum phishing detection framework, integrating research-driven feature engineering and model development to achieve a reliable and performance-oriented solution.
EA4 Consequences for Society and the Environment	The proposed phishing detection framework can contribute to reducing cybercrime, credential theft, financial fraud and unauthorized data access by improving phishing email detection accuracy. Organizations, cybersecurity professionals and digital service providers may use the system to strengthen digital security infrastructure and improve user protection against evolving phishing attacks.

CHAPTER 6

DISCUSSION AND CONCLUSION

6.1 Implications and Contributions

This thesis presents a hybrid classical–quantum machine learning framework for email phishing detection, integrating a classical Support Vector Machine (SVM) and a Quantum Support Vector Machine (QSVM) within a stacking ensemble architecture. The results demonstrate that quantum-enhanced kernel methods can be meaningfully combined with classical learning models to improve classification robustness in a real-world cybersecurity task. The following specific contributions are made:

1. Demonstration of practical QSVM deployment under NISQ constraints. A persistent challenge in Quantum Machine Learning (QML) research is the gap between theoretical quantum advantage and practical deployment on near-term hardware. This thesis addresses this directly by designing a batch-wise training strategy that reduces each kernel matrix from an infeasible $\mathcal{O}(N^2 \cdot 2^d)$ computation on the full dataset to tractable 500×500 blocks, enabling QSVM training on a standard laptop without access to quantum hardware. The resulting system achieves 98.85% accuracy which is a statistically significant improvement over the classical SVM baseline ($p = 0.018$, McNemar’s test) demonstrating that quantum kernel methods are deployable in practical settings today, not only in idealized conditions.

2. Quantified quantum advantage in phishing detection. The most practically significant finding of this study is the QSVM’s reduction of false negatives from 25 (SVM) to 5 which is an 80% improvement in missed phishing detection without meaningfully degrading precision. This result suggests that the ZZ entanglement interactions encoded by the `ZZFeatureMap` capture cross-feature correlations relevant to phishing detection that are inaccessible to the classical RBF kernel. Specifically, the six selected features spanning JavaScript obfuscation, content entropy and domain trust appear to exhibit non-linear joint dependencies that quantum feature maps are well-positioned to exploit.

3. A hybrid dataset combining email and web-level phishing signals. A novel hybrid dataset was constructed by merging email metadata with web-level content features,

augmented by 24 engineered features covering structural URL characteristics, entropy measures, domain trust, semantic coherence and brand consistency. This dataset provides a richer and more realistic representation of phishing behaviour than either source alone and is used uniformly across all models to ensure fair comparison. The feature engineering pipeline is fully documented and reproducible.

4. Principled ensemble integration of classical and quantum outputs. The stacking meta-classifier resolves the precision–recall trade-off between the SVM (low false positives, higher false negatives) and the QSVM (very low false negatives, slightly higher false positives), achieving 9 false positives and 6 false negatives simultaneously. Although the ensemble’s improvement over the QSVM alone is not statistically significant ($p = 0.646$), it provides a consistent and reliable system that inherits the best error profile of both base models which is a practically valuable property in a security deployment.

5. Comprehensive statistical and ablation validation. The framework is assessed not only using the standard classification measures, but also using a three-axis ablation study on component selection, probability weighting and meta-classifier regularisation, along with the McNemar’s test, the paired t-test and Wilcoxon signed-rank test and Cohen’s d effect size analysis. This level of rigour supports evidence-based conclusions about the sources and limits of the observed performance gains.

6.2 Limitations

The following limitations of this thesis are acknowledged:

- **Quantum simulation rather than real hardware.** All quantum computations were performed in an idealized, noise-free simulation environment, `qiskit.primitives.Sampler`, rather than on physical quantum hardware. Consequently, the effects of hardware noise, decoherence and gate errors were not evaluated and the practical performance of the QSVM on real quantum devices remains unverified.
- **Severe computational overhead of quantum kernel simulation.** QSVM training required approximately 4.6 hours across 13 batches of 500 samples, compared to 2.8 seconds for the classical SVM — a speed ratio of up to $6,480\times$. This cost is inherent to classical simulation of quantum circuits and scales exponentially with qubit count. As a result, the system cannot be trained incrementally, retrained on

new phishing patterns in real time or scaled to substantially larger datasets without access to quantum hardware or more efficient simulation backends.

- **Restricted to six features due to qubit budget.** The `ZZFeatureMap` allocates one qubit per input feature, meaning the quantum circuit complexity scales directly with feature dimensionality. Under classical simulation constraints, only six features could be encoded, selected via Mutual Information ranking. A richer feature encoding strategy or a more expressive feature map could potentially capture additional discriminative signal.
- **Fixed feature map architecture.** The `ZZFeatureMap` with one repetition and linear entanglement was used throughout this study without systematic exploration of alternative circuit architectures. Different feature maps such as `PauliFeatureMap` with higher-order interactions, or custom ansatz circuits may encode feature relationships more effectively for this specific dataset and could yield meaningfully different kernel matrices and classification outcomes.
- **Dataset scope and recency.** The dataset was constructed from two publicly available sources, the CEAS_08 email corpus and the Malicious and Benign Webpage Dataset, both of which reflect phishing patterns from a specific historical period. The dataset may not capture zero-day phishing techniques, adversarial phishing campaigns designed to evade detection or emerging attack vectors such as QR code phishing or AI-generated phishing content.

6.3 Future Work

The following directions are identified for extending this work:

- **Deployment on real quantum hardware.** The most direct extension of this work is to execute the QSVM kernel on a physical quantum processor such as IBM Quantum or IonQ, rather than classical simulation. This would quantify the impact of hardware noise on kernel quality and classification performance and would determine whether the quantum advantage observed under ideal simulation survives in a realistic hardware setting. Noise mitigation techniques such as zero-noise extrapolation or probabilistic error cancellation could be explored to preserve kernel fidelity under decoherence.

- **Custom and higher-expressibility feature maps.** The `ZZFeatureMap` used in this study is a standard first-order entangling circuit with limited expressibility for complex feature interactions. Future work should systematically evaluate alternative quantum feature maps, including:
 - `PauliFeatureMap` with higher-order Pauli interactions beyond `ZZ`;
 - Trainable quantum kernels where the feature map parameters are optimised end-to-end via gradient descent on a kernel-alignment objective;
 - Data re-uploading circuits that encode features multiple times across circuit layers to increase representational capacity without increasing qubit count.

These approaches may enable richer kernel representations while remaining feasible within the same 6-qubit budget.

- **Encoding more features within the qubit budget.** Rather than selecting only 6 features to match the qubit count, future work could explore amplitude encoding or basis encoding schemes that represent multiple features per qubit. Alternatively, dimensionality reduction techniques such as PCA or autoencoders could be applied prior to quantum encoding to compress the full 36-feature space into a 6-dimensional quantum-compatible representation that preserves more information than MI-based univariate selection.
- **Expanding and diversifying the dataset.** The framework should be evaluated on more recent and diverse phishing datasets that capture contemporary attack patterns, including AI-generated phishing emails, multi-stage phishing campaigns and mobile-targeted phishing. Cross-dataset evaluation that is training on one dataset and testing on another would also provide a stronger measure of generalization capability.
- **Regularisation and hyperparameter optimisation.** The ablation study demonstrated that the ensemble accuracy improves monotonically with the meta-classifier’s regularisation parameter C , reaching 0.9942 at $C = 100.0$ compared to 0.9904 at $C = 1.0$. A principled hyperparameter search using nested cross-validation to avoid test set leakage could identify the optimal C value and potentially other QSVM hyperparameters such as the feature map repetition count (`reps`) and entanglement topology, leading to further performance improvements without any architectural changes.

6.4 Conclusion

This thesis investigated whether quantum kernel methods can provide a meaningful and measurable advantage over classical machine learning for phishing detection and whether a hybrid ensemble can combine the strengths of both models into a single, reliable system.

Compared with the classical SVM baseline, the proposed framework achieves a 1.21% improvement in accuracy and significantly reduces false negatives by 80%, demonstrating the benefit of integrating quantum kernel learning within a hybrid ensemble-based phishing detection system.

The results demonstrate that the Quantum Support Vector Machine (QSVM) perform well than classical SVM in most of the evaluation metrics. In particular, QSVM achieves higher recall (0.993 vs 0.967) and higher overall accuracy (0.9885 vs 0.9783), indicating its stronger ability to correctly identify phishing emails while reducing missed detections.

The stacking ensemble further improves the performance, with an accuracy of 0.9904 and the best F1 score of 0.990. It also maintains strong recall (0.992), showing a better balance between precision and recall compared to the individual models. The performance boost provided by the ensemble layer is, however, marginal in comparison to that of QSVM alone, suggesting that the boost is mostly due to the quantum model and not the ensemble layer.

The overall results of this thesis indicate that the quantum-enhanced models can provide measurable improvements over classical models in the phishing detection task and that ensemble learning can help to further stabilize the detection performance.

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APPENDIX A

THESIS SELF-ASSESSMENT REPORT

- **Title of the Research:** Batch-Wise Ensemble Model using Classical and Quantum SVM for Email Phishing Detection
- **Name of Supervisor:** Dr. Fernaz Narin Nur
- **Abstract of the Research:** Email phishing is a very serious form of attack in today's communication environment where criminals send fraudulent e-mails to obtain sensitive information including login names, financial information and personal data. Traditional rule-based and blacklist-based solutions are still inadequate to defend against the more advanced phishing attacks, as they continually update their rules and blacklists. While classical machine learning algorithms and learning patterns from data have been attempted to detect attacks, there are problems because of the reliance on feature engineering and generalization across the attack types. Recently Quantum Machine Learning (QML) has emerged as a new paradigm of Machine Learning that has been demonstrated to improve the accuracy of interactions with classical data using quantum entanglement to encode data into high dimensional Hilbert spaces and quantum kernel methods. This thesis suggests the construction of a hybrid dataset by modelling the attributes of real world phishing behavior and some of the raw dataset attributes. The use of this data is proposed using a combination of Support Vector Machine (SVM) and Quantum Support Vector Machine (QSVM) in a single ensemble architecture, called hybrid classical-quantum machine learning. Classical SVM has good deterministic classification ability, but the ability of separability of QSVM is strengthened by quantum feature mapping using the kernel method in a feature space of higher dimension. In both models, both outputs are calibrated to ensure the same probabilistic interpretation when using ensemble learning and for comparing both models in a common framework. There are detailed experiments conducted using statistical evaluation methods, as well as ablation experiments to determine the system and observe the contribution of the individual parts. The experiments show that QSVM is able to reduce the number of false negatives by 80% while maintaining the same precision compared to the classical SVM. Overall accuracy of 99.04% outperforms both of

the base models and overcomes the issue of precision – recall trade-off between the two base models. Overall, this work highlights the effectiveness of combining multi-source feature engineering with classical quantum hybrid learning to enhance phishing email detection systems and improve cybersecurity resilience.

Mapping of Attained COs

Table A.1: Mapping of Attained COs

Term	COs	CO Statements	POs	Level	Relevant Chapter/Section
Spring	CO1	Identify a real-life problem / contemporary issues that can be translated into an engineering or computing solution through design, development and validation with focus on research and innovation contribution.	PO12	H	1.1, 2.2
Spring	CO2	Identify functional requirements and system outcomes considering software/hardware specifications and standards.	PO2	H	1.2, 3.2
Spring	CO3	Identify sub-components of a complex problem and plan timeline and budget using project management skills.	PO11	H	1.3
Spring	CO4	Investigate and analyze engineering/computing systems using given specifications and requirements.	PO4	H	4.1, 4.2, 4.3
Spring	CO5	Design, build and evaluate the proposed solution based on system requirements.	PO3	H	3.5,3.6,3.7

Term	COs	CO Statements	POs	Level	Relevant Chapter/Section
Fall	CO1	Use modern analysis and design tools for system and sub-system validation.	PO5	M	4.1, 4.2, 4.3
Fall	CO2	Assess social impacts and responsibilities of engineering design projects.	PO6	M	5.1.3, 5.1.4
Fall	CO3	Evaluate ethical and professional implications of the proposed solution.	PO8	H	5.1.1
Fall	CO4	Identify and evaluate environmental impact and sustainability of the system.	PO7	L	5.1.2, 5.1.4
Fall	CO5	Function effectively as a team in project development activities.	PO9	M	5.4
Fall	CO6	Present design results through written technical documents and oral presentations.	PO10	H	3, 4

Assesment Question and Answer

- What real-world problem did you address, what research gap did you identify and how does your solution contribute to future innovation or advancement in the field?**

This project targets the ever increasing danger of phishing e-mail which is still one of the most frequent means of attacks. Current phishing detection techniques are mostly based on classical machine learning methods and typically use email-based features or URL-based features, separately. Limited research was found on the application of quantum machine learning in the detection of phishing and the absence of integrating hybrid datasets that incorporate both the email and web- page features. To fill this gap, a hybrid quantum–classical phishing detection framework

was developed.

2. How did you analyze the problem requirements, identify constraints and determine the functional specifications of your proposed solution?

The problem requirements were identified by reviewing the phishing detection literature and analyzing the current cybersecurity challenges. There were a number of key functional requirements, such as compatibility with classical and quantum models, the ability to handle different types of features and the ability to classify phishing. The most significant challenges were the limited supply of qubits, the computational complexity of quantum kernel evaluations and the fact that quantum computers need to be executed using a simulator and have limited natural resources. These considerations led to the design of a scalable hybrid framework that trained the QSVM in a batch-wise manner and made predictions using an ensemble of them

3. How did you plan, schedule, allocate resources and manage risks throughout the project to ensure successful completion?

The project was divided into sequential phases including literature review, dataset construction, feature engineering, preprocessing, model development, experimentation, statistical validation and thesis writing. Computational resources were allocated efficiently by using classical simulation environments and optimized machine learning libraries. Risks such as excessive QSVM memory requirements and training failures were mitigated through feature selection, dimensionality reduction, batch-wise training and continuous experimental validation.

4. What investigation and analysis methods did you use to validate your assumptions and how did the results influence your solution?

The proposed framework was validated through experimental evaluation using accuracy, precision, recall, F1-score, ROC-AUC, confusion matrices, statistical significance testing and ablation studies. McNemar's test, paired statistical tests and effect size analysis were used to assess whether observed performance differences were statistically meaningful. The results confirmed the effectiveness of quantum kernel methods and demonstrated that the ensemble model provided the most balanced overall performance.

5. How did you design, implement and evaluate your solution to ensure that it satisfies the identified requirements and constraints?

The solution was designed as a hybrid framework consisting of classical SVM, QSVM and a stacking ensemble classifier. A feature-engineered hybrid dataset was constructed by combining email and webpage information. The framework was implemented using Python, Scikit-learn and Qiskit. Performance was evaluated using a stratified train–test split and multiple classification metrics. Batch-wise QSVM training ensured that computational constraints were satisfied while maintaining model effectiveness.

6. Which modern engineering or computing tools did you use, why were they selected and how did they support the development and validation of your project?

The primary development language was chosen as Python, which has a rich ecosystem of machine learning. Data preprocessing, feature selection, implementation of classical SVM, ensemble learning and the assessment of the learning processes were performed using scikit-learn. The quantum feature map and the QSVM models were implemented using Qiskit. Pandas and NumPy were used for data manipulation and numerical analysis and Matplotlib was used for visualization of results. These tools allowed efficient development, experiments and validation of the proposed framework.

7. What are the societal impacts of your project and how does your solution address the needs and responsibilities of stakeholders?

The project helps improve cyber security by lowering the chances of phishing attacks being successful. Better phishing detection is essential for safeguarding against financial loss, data breaches and identity theft for individuals, organizations and critical digital infrastructure. The framework can help stakeholders to improve the reliability of detection and to continue research into the new technologies of quantum applications in cybersecurity.

8. What ethical and professional issues are associated with your project and how did you address them during the design and implementation process?

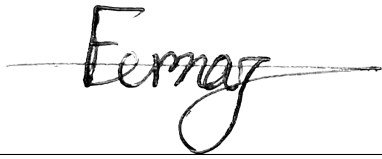
Ethical considerations include responsible handling of cybersecurity data, ensuring transparency of experimental procedures and avoiding misuse of detection technologies. All publicly available datasets were utilized and all experiments were made only for defensive cyber security purposes. The results, limitations and assumptions were presented honestly to assure the integrity and repeatability of scientific research.

9. How does your project consider environmental sustainability and what measures were taken to minimize negative environmental impacts?

The project was conducted entirely through software-based experimentation using existing computational resources, eliminating the need for additional physical infrastructure. Computational costs were reduced through feature selection and batch-wise training strategies. By improving phishing detection efficiency, the framework also contributes indirectly to sustainable and secure digital systems.

10. How would you communicate your project's objectives, methodology, results and significance to both technical and non-technical audiences?

The project can be explained to the technical audience through detailed description of the process of building the data set, quantum feature mapping, model building, experimental implementation methodology and statistical validation. The project can be described to non-technical audiences as an innovative, high-tech solution that uses a mixture of classic artificial intelligence and cutting-edge quantum computing technology to better detect and identify phishing emails and to better establish cybersecurity.



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